Distance matrices, Gram matrices and multidimensional scaling (MDS)

MATH 2740 – Mathematics of Data Science – Unit U07

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The University of Manitoba campuses are located on original lands of Anishinaabeg, Ininew, Anisininew, Dakota and Dene peoples, and on the National Homeland of the Red River Métis. We respect the Treaties that were made on these territories, we acknowledge the harms and mistakes of the past, and we dedicate ourselves to move forward in partnership with Indigenous communities in a spirit of Reconciliation and collaboration.

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Gram matrices and Euclidean distance matrices

The centring matrix as an orthogonal projector

Classical multidimensional scaling (MDS)

Algebraic duality between PCA and classical MDS

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From coordinates to distances and back

In principal component analysis (Unit U05), we start with a data matrix $X \in \mathcal{M}_{np}(\mathbb{R})$ containing the coordinates of n observations $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n \in \mathbb{R}^p$ and project them onto a lower-dimensional subspace $\mathbb{R}^k$ ($k \leq p$)

From the coordinates $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n \in \mathbb{R}^p$, it is straightforward to compute all pairwise Euclidean distances

$$d_{ij} = \|\boldsymbol{x}_i - \boldsymbol{x}_j\| = \sqrt{(\boldsymbol{x}_i - \boldsymbol{x}_j) \bullet (\boldsymbol{x}_i - \boldsymbol{x}_j)}, \qquad i, j = 1, \ldots, n$$

The inverse problem: what if we do *not* know the coordinates X, and only have a table of pairwise distances or dissimilarities $D = [d_{ij}] \in \mathcal{M}_n(\mathbb{R})$ between n objects?

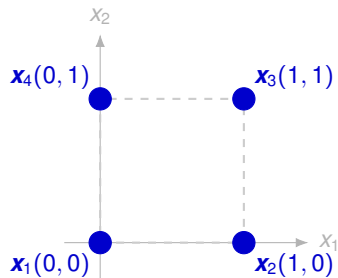
Running example: four points on a square

Consider $n = 4$ points in $p = 2$ dimensions forming a unit square (in counterclockwise order):

$$\boldsymbol{x}_1 = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \qquad \boldsymbol{x}_2 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad \boldsymbol{x}_3 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \qquad \boldsymbol{x}_4 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

The coordinate matrix $X \in \mathcal{M}_{42}(\mathbb{R})$ is:

$$X = \begin{pmatrix} \boldsymbol{x}_1^T \\ \boldsymbol{x}_2^T \\ \boldsymbol{x}_3^T \\ \boldsymbol{x}_4^T \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{pmatrix}$$



Running example: distance matrix of the square

Pairwise Euclidean distances $d_{ij} = \|\boldsymbol{x}_i - \boldsymbol{x}_j\|$ between the vertices are easy to compute here:

- **Adjacent vertices (edges of length 1)**:

$$d_{12} = d_{23} = d_{34} = d_{14} = 1$$

- **Opposite vertices (diagonals of length $\sqrt{2}$)**:

$$d_{13} = d_{24} = \sqrt{1^2 + 1^2} = \sqrt{2}$$

- **Self-distances**: $d_{ii} = 0$ for $i = 1, \ldots, 4$

This gives the pairwise distance matrix $D \in \mathcal{M}_4(\mathbb{R})$:

$$D = \begin{pmatrix} 0 & 1 & \sqrt{2} & 1 \\ 1 & 0 & 1 & \sqrt{2} \\ \sqrt{2} & 1 & 0 & 1 \\ 1 & \sqrt{2} & 1 & 0 \end{pmatrix}$$

Running example: reconstructing X from D

The fundamental question of this unit, formulated on our running example:

Given the distance matrix:

$$D = \begin{pmatrix} 0 & 1 & \sqrt{2} & 1 \\ 1 & 0 & 1 & \sqrt{2} \\ \sqrt{2} & 1 & 0 & 1 \\ 1 & \sqrt{2} & 1 & 0 \end{pmatrix}$$

Can we reconstruct coordinates:

$$X = \begin{pmatrix} 0 & 0 \\ 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{pmatrix} ?$$

- Can we deduce from D alone that the points embed in dimension $p = 2$?
- Can we recover the vertices of the square (up to translation, rotation and reflection)?
- What mathematical operations take us from the distance matrix D to the coordinates X?

Where do distance-only datasets arise?

In many scientific applications, objects do not come with explicit coordinates in $\mathbb{R}^p$, or only pairwise comparisons are measured:

- **Geography and transport**: road distances, flight distances or travel times between n cities
- **Genomics and evolutionary biology**: sequence edit distances or evolutionary divergence times between n species
- **Psychology and sensory science**: human judgements of dissimilarity between colours, sounds, faces or products
- **Cheminformatics and text analysis**: structural dissimilarity between molecules or cosine/Jaccard distances between documents

The goal of multidimensional scaling (MDS)

Given an $n \times n$ matrix $D = [d_{ij}]$ of pairwise distances between n objects:

1. **Exact Euclidean embedding**: do there exist points $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n \in \mathbb{R}^p$ (for some dimension p) such that

$$\|\boldsymbol{x}_i - \boldsymbol{x}_j\| = d_{ij} \qquad \forall i, j = 1, \ldots, n?$$

2. **Dimensionality reduction and visualisation**: for a chosen small dimension k (typically $k = 1, 2$ or 3), find coordinates $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n \in \mathbb{R}^k$ whose pairwise Euclidean distances $\|\boldsymbol{x}_i - \boldsymbol{x}_j\|$ best approximate d_{ij}

This family of methods is called **multidimensional scaling** (**MDS**). When D is treated as a Euclidean distance matrix and coordinates are recovered via spectral decomposition, it is called **classical MDS** (or **principal coordinate analysis**, **PCoA**)

Non-uniqueness: rigid motions preserve distances

Before trying to recover coordinates $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n \in \mathbb{R}^p$ from pairwise distances $d_{ij} = \|\boldsymbol{x}_i - \boldsymbol{x}_j\|$, observe that coordinates can *never* be unique!

Proposition 95 (Invariance of Euclidean distances under rigid motions)

Let $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n \in \mathbb{R}^p$, let $\boldsymbol{s} \in \mathbb{R}^p$ be a translation vector and let $Q \in \mathcal{M}_p(\mathbb{R})$ be an orthogonal matrix ($Q^T Q = \mathbb{I}_p$). Define transformed points

$$\boldsymbol{y}_i = Q\boldsymbol{x}_i + \boldsymbol{s}, \qquad i = 1, \ldots, n$$

Then all pairwise Euclidean distances are preserved:

$$\|\boldsymbol{y}_i - \boldsymbol{y}_j\| = \|\boldsymbol{x}_i - \boldsymbol{x}_j\| \qquad \forall i, j = 1, \ldots, n$$

Thus, distances only determine a point configuration up to **translation**, **rotation** and **reflection** (collectively called **rigid motions** or **Euclidean isometries**)

Proof of Proposition 95

Let $\boldsymbol{y}_i = Q\boldsymbol{x}_i + \boldsymbol{s}$ for $i = 1, \ldots, n$, where $\boldsymbol{s} \in \mathbb{R}^p$ and $Q^T Q = \mathbb{I}_p$

First, taking the difference between $\boldsymbol{y}_i$ and $\boldsymbol{y}_j$ eliminates the translation vector $\boldsymbol{s}$:

$$\boldsymbol{y}_i - \boldsymbol{y}_j = (Q\boldsymbol{x}_i + \boldsymbol{s}) - (Q\boldsymbol{x}_j + \boldsymbol{s}) = Q\boldsymbol{x}_i - Q\boldsymbol{x}_j = Q(\boldsymbol{x}_i - \boldsymbol{x}_j)$$

Next, compute the squared Euclidean norm using the orthogonality of Q:

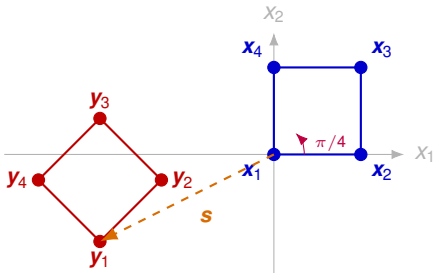
$$\begin{aligned} \|\boldsymbol{y}_i - \boldsymbol{y}_j\|^2 &= \|Q(\boldsymbol{x}_i - \boldsymbol{x}_j)\|^2 \\ &= \big(Q(\boldsymbol{x}_i - \boldsymbol{x}_j)\big)^T \big(Q(\boldsymbol{x}_i - \boldsymbol{x}_j)\big) \\ &= (\boldsymbol{x}_i - \boldsymbol{x}_j)^T Q^T Q(\boldsymbol{x}_i - \boldsymbol{x}_j) \\ &= (\boldsymbol{x}_i - \boldsymbol{x}_j)^T \mathbb{I}_p (\boldsymbol{x}_i - \boldsymbol{x}_j) = \|\boldsymbol{x}_i - \boldsymbol{x}_j\|^2 \end{aligned}$$

Taking the non-negative square root yields $\|\boldsymbol{y}_i - \boldsymbol{y}_j\| = \|\boldsymbol{x}_i - \boldsymbol{x}_j\|$ □

Running example: rigid motion of the square

Apply a rigid motion $\boldsymbol{y}_i = Q\boldsymbol{x}_i + \boldsymbol{s}$ to our four vertices, with rotation counterclockwise by $\theta = \pi/4$ and translation down and left:

$$Q = \begin{pmatrix} \cos(\pi/4) & -\sin(\pi/4) \\ \sin(\pi/4) & \cos(\pi/4) \end{pmatrix} = \frac{\sqrt{2}}{2}\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}, \qquad \boldsymbol{s} = \begin{pmatrix} -2 \\ -1 \end{pmatrix}$$



Rigid motion: $\boldsymbol{y}_i = Q\boldsymbol{x}_i + \boldsymbol{s}$

- Original square X
- Transformed square Y

Distances preserved:

$\|\boldsymbol{y}_i - \boldsymbol{y}_j\| = \|\boldsymbol{x}_i - \boldsymbol{x}_j\|$

$D_Y = D$

Running example: computing transformed coordinates

Apply the transformation $\boldsymbol{y}_i = Q\boldsymbol{x}_i + \boldsymbol{s}$ to each vertex of the square:

- **Vertex** $\boldsymbol{x}_1 = (0,0)^T$:

$$\boldsymbol{y}_1 = Q\boldsymbol{x}_1 + \boldsymbol{s} = Q\begin{pmatrix}0\\0\end{pmatrix} + \begin{pmatrix}-2\\-1\end{pmatrix} = \begin{pmatrix}-2\\-1\end{pmatrix}$$

- **Vertex** $\boldsymbol{x}_2 = (1,0)^T$ **(in detail)**:

$$\boldsymbol{y}_2 = Q\boldsymbol{x}_2 + \boldsymbol{s} = \frac{\sqrt{2}}{2}\begin{pmatrix}1 & -1\\1 & 1\end{pmatrix}\begin{pmatrix}1\\0\end{pmatrix} + \begin{pmatrix}-2\\-1\end{pmatrix} = \begin{pmatrix}\frac{\sqrt{2}}{2} - 2\\\frac{\sqrt{2}}{2} - 1\end{pmatrix} \approx \begin{pmatrix}-1.293\\-0.293\end{pmatrix}$$

- **Remaining vertices** $\boldsymbol{x}_3 = (1,1)^T$ **and** $\boldsymbol{x}_4 = (0,1)^T$ **(briefly)**:

$$\boldsymbol{y}_3 = Q\boldsymbol{x}_3 + \boldsymbol{s} = \begin{pmatrix}0\\\sqrt{2}\end{pmatrix} + \begin{pmatrix}-2\\-1\end{pmatrix} = \begin{pmatrix}-2\\\sqrt{2} - 1\end{pmatrix} \approx \begin{pmatrix}-2\\0.414\end{pmatrix}$$

$$\boldsymbol{y}_4 = Q\boldsymbol{x}_4 + \boldsymbol{s} = \begin{pmatrix}-\frac{\sqrt{2}}{2}\\\frac{\sqrt{2}}{2}\end{pmatrix} + \begin{pmatrix}-2\\-1\end{pmatrix} = \begin{pmatrix}-\frac{\sqrt{2}}{2} - 2\\\frac{\sqrt{2}}{2} - 1\end{pmatrix} \approx \begin{pmatrix}-2.707\\-0.293\end{pmatrix}$$

The resulting configuration Y has the **exact same** pairwise distance matrix D as X

Removing translation ambiguity by centring at the origin

Because shifting all points by a constant vector $\boldsymbol{s} \in \mathbb{R}^p$ leaves every pairwise distance unchanged, we must fix a reference location to obtain a well-defined reconstruction

Convention in classical MDS: we require the reconstructed points $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n \in \mathbb{R}^p$ to have their **centroid** (sample mean) at the origin $\mathbf{0} \in \mathbb{R}^p$:

$$\bar{\boldsymbol{x}} = \frac{1}{n} \sum_{i=1}^{n} \boldsymbol{x}_i = \mathbf{0} \iff \sum_{i=1}^{n} \boldsymbol{x}_i = \mathbf{0}$$

- Fixing $\sum_{i=1}^{n} \boldsymbol{x}_i = \mathbf{0}$ completely removes the translation ambiguity
- Any remaining ambiguity is purely an orthogonal transformation $Q \in \mathcal{M}_p(\mathbb{R})$ (rotation or reflection about the origin)
- Algebraically, why does fixing the origin help? Because once an origin is fixed, **distances determine inner products**, and inner products give us a **Gram matrix** that we can diagonalise!

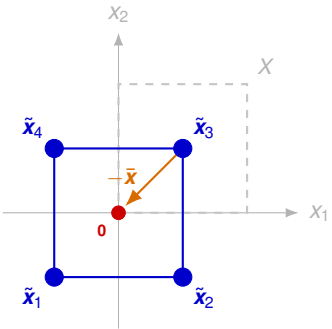
Running example: centring the square at the origin

The centroid (sample mean) of the original four points is:

$$\bar{\boldsymbol{x}} = \frac{1}{4}\sum_{i=1}^{4} \boldsymbol{x}_i = \frac{1}{4}\begin{pmatrix} 0+1+1+0 \\ 0+0+1+1 \end{pmatrix} = \begin{pmatrix} 1/2 \\ 1/2 \end{pmatrix}$$

Subtracting $\bar{\boldsymbol{x}}$ centres the points at the origin, giving the centred data matrix

$$\widetilde{X} = \begin{pmatrix} -1/2 & -1/2 \\ 1/2 & -1/2 \\ 1/2 & 1/2 \\ -1/2 & 1/2 \end{pmatrix} \in \mathcal{M}_{42}(\mathbb{R})$$



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The squared Euclidean distance matrix

Data matrix convention: observations in rows

Throughout this unit, we represent a configuration of n points in $\mathbb{R}^p$ by a data matrix $X \in \mathcal{M}_{np}(\mathbb{R})$ whose ith row is the transpose of the column vector $\boldsymbol{x}_i \in \mathbb{R}^p$:

$$X = \begin{pmatrix} \boldsymbol{x}_1^T \\ \boldsymbol{x}_2^T \\ \vdots \\ \boldsymbol{x}_n^T \end{pmatrix} = \begin{pmatrix} x_{11} & x_{12} & \cdots & x_{1p} \\ x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{np} \end{pmatrix} \in \mathcal{M}_{np}(\mathbb{R})$$

- Rows $i = 1, \ldots, n$ are **observations** (objects, samples, cities or individuals)
- Columns $j = 1, \ldots, p$ are **features** (also called variables or coordinate axes)

The Gram matrix of a set of vectors

Definition 96 (Gram matrix)

Let $X \in \mathcal{M}_{np}(\mathbb{R})$ have rows $\boldsymbol{x}_1^T, \ldots, \boldsymbol{x}_n^T \in \mathbb{R}^p$. The ***Gram matrix*** *(or* ***inner product matrix****) of $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n$ is the $n \times n$ matrix*

$$G = XX^T \in \mathcal{M}_n(\mathbb{R})$$

whose (i, j) entry is the standard inner product between $\boldsymbol{x}_i$ and $\boldsymbol{x}_j$:

$$g_{ij} = \boldsymbol{x}_i \bullet \boldsymbol{x}_j = \boldsymbol{x}_i^T \boldsymbol{x}_j = \sum_{m=1}^{p} x_{im} x_{jm}, \qquad i, j = 1, \ldots, n$$

Explicitly, in block row-times-column form:

$$G = XX^T = \begin{pmatrix} \boldsymbol{x}_1^T \\ \vdots \\ \boldsymbol{x}_n^T \end{pmatrix} \begin{pmatrix} \boldsymbol{x}_1 & \cdots & \boldsymbol{x}_n \end{pmatrix} = \begin{pmatrix} \boldsymbol{x}_1^T\boldsymbol{x}_1 & \boldsymbol{x}_1^T\boldsymbol{x}_2 & \cdots & \boldsymbol{x}_1^T\boldsymbol{x}_n \\ \boldsymbol{x}_2^T\boldsymbol{x}_1 & \boldsymbol{x}_2^T\boldsymbol{x}_2 & \cdots & \boldsymbol{x}_2^T\boldsymbol{x}_n \\ \vdots & \vdots & \ddots & \vdots \\ \boldsymbol{x}_n^T\boldsymbol{x}_1 & \boldsymbol{x}_n^T\boldsymbol{x}_2 & \cdots & \boldsymbol{x}_n^T\boldsymbol{x}_n \end{pmatrix}$$

Geometric meaning of the Gram matrix

Every entry of the Gram matrix $G = XX^T$ encodes geometric information relative to the origin **0**:

- **Diagonal entries** give squared lengths (squared distances from the origin):

$$g_{ii} = \boldsymbol{x}_i^T \boldsymbol{x}_i = \|\boldsymbol{x}_i\|^2 \geq 0, \qquad i = 1, \ldots, n$$

- **Off-diagonal entries** give lengths and angles θ_{ij} between vectors $\boldsymbol{x}_i$ and $\boldsymbol{x}_j$:

$$g_{ij} = \boldsymbol{x}_i^T \boldsymbol{x}_j = \|\boldsymbol{x}_i\| \, \|\boldsymbol{x}_j\| \cos\theta_{ij}, \qquad i \neq j$$

- **Orthogonal invariance**: if we rotate or reflect all points by an orthogonal matrix $Q \in \mathcal{M}_p(\mathbb{R})$ (so $\boldsymbol{x}_i \mapsto Q\boldsymbol{x}_i$ and $X \mapsto XQ^T$), the Gram matrix is **unchanged**:

$$(XQ^T)(XQ^T)^T = XQ^TQX^T = X\mathbb{I}_pX^T = XX^T = G$$

Running example: Gram matrix of the square

Recall our four points $X \in \mathcal{M}_{42}(\mathbb{R})$:

$$\boldsymbol{x}_1 = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \qquad \boldsymbol{x}_2 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad \boldsymbol{x}_3 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \qquad \boldsymbol{x}_4 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

The Gram matrix $G = XX^T \in \mathcal{M}_4(\mathbb{R})$ is:

$$G = XX^T = \begin{pmatrix} 0 & 0 \\ 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & 1 \end{pmatrix}$$

Geometric interpretation of the entries:

- **Diagonal entries** $g_{ii} = \|\boldsymbol{x}_i\|^2$: squared distances from $\mathbf{0}$ are $(0, 1, 2, 1)$
- **First row/column**: all zero because $\boldsymbol{x}_1 = \mathbf{0}$ is at the origin
- **Orthogonality**: $g_{24} = 0$ because $\boldsymbol{x}_2 = (1, 0)^T \perp \boldsymbol{x}_4 = (0, 1)^T$ ($\theta_{24} = \pi/2$)
- **Angle**: $g_{23} = 1$ because $\|\boldsymbol{x}_2\|\|\boldsymbol{x}_3\| \cos(\pi/4) = 1 \cdot \sqrt{2} \cdot \frac{\sqrt{2}}{2} = 1$

Positive semi-definite and positive definite matrices

Definition 97 (Positive semi-definite and positive definite matrices)

Let $A \in \mathcal{M}_n(\mathbb{R})$ be a symmetric matrix ($A^T = A$).

- *A is **positive semi-definite** (**PSD**) if*

$$\boldsymbol{v}^T A \boldsymbol{v} \geq 0 \qquad \forall \boldsymbol{v} \in \mathbb{R}^n$$

- *A is **positive definite** (**PD**) if*

$$\boldsymbol{v}^T A \boldsymbol{v} > 0 \qquad \forall \boldsymbol{v} \in \mathbb{R}^n \setminus \{\mathbf{0}\}$$

Recall from Theorem **??** (Unit U04) that all eigenvalues of a real symmetric matrix $A \in \mathcal{M}_n(\mathbb{R})$ are real. Equivalently

- A is positive semi-definite $\iff$ all eigenvalues of A satisfy $\lambda_i \geq 0$

- A is positive definite $\iff$ all eigenvalues of A satisfy $\lambda_i > 0$ (in which case A is invertible)

You have already encountered these matrices!

Even without the name, we have already proved and used these properties for matrices of the form A^TA and AA^T (with $A \in \mathcal{M}_{np}(\mathbb{R})$):

- **Unit U02 (Theorem 25)**: we proved that for any $A \in \mathcal{M}_{np}(\mathbb{R})$, both $AA^T \in \mathcal{M}_n(\mathbb{R})$ and $A^TA \in \mathcal{M}_p(\mathbb{R})$ are **symmetric**

- **Unit U03 (Least squares, invertibility of A^TA)**: for any $\boldsymbol{x} \in \mathbb{R}^p$, we wrote

$$\boldsymbol{x}^T(A^TA)\boldsymbol{x} = (A\boldsymbol{x})^T(A\boldsymbol{x}) = \|A\boldsymbol{x}\|^2 \geq 0$$

  which says precisely that A^TA is **positive semi-definite**! When A has linearly independent columns ($A\boldsymbol{x} = \boldsymbol{0} \iff \boldsymbol{x} = \boldsymbol{0}$), $\|A\boldsymbol{x}\|^2 > 0$ for all $\boldsymbol{x} \neq \boldsymbol{0}$, so A^TA is **positive definite** (and hence invertible)

- **Unit U04 (SVD, Theorem ??)**: we proved that all eigenvalues of $A^T A$ and AA^T are **real and non-negative** ($\lambda_i \geq 0$), which allowed us to define singular values $\sigma_i = \sqrt{\lambda_i}$

- **Unit U05 (PCA)**: the sample covariance matrix $S_X = \frac{1}{n-1} \widetilde{X}^T \widetilde{X}$ is symmetric positive semi-definite, and its eigenvalues $\lambda_i \geq 0$ are the variances along the principal axes

Fundamental properties of Gram matrices

Notice that for a data matrix $X \in \mathcal{M}_{np}(\mathbb{R})$, its Gram matrix $G = XX^T \in \mathcal{M}_n(\mathbb{R})$ is simply the matrix AA^T from the SVD in Unit U04 (with $A = X$), whose eigenvectors are the left singular vectors of X

Theorem 98 (Properties of the Gram matrix)

Let $X \in \mathcal{M}_{np}(\mathbb{R})$ and let $G = XX^T \in \mathcal{M}_n(\mathbb{R})$ be its Gram matrix. Then:

1. *G is symmetric and positive semi-definite, with real non-negative eigenvalues $\lambda_1 \geq \cdots \geq \lambda_n \geq 0$*
2. $\text{rank}(G) = \text{rank}(X) \leq \min(n, p)$
3. $\text{tr}(G) = \sum_{i=1}^{n} \|\boldsymbol{x}_i\|^2$

Characterisation of Gram matrices via spectral factorisation

In the SVD (Theorems **??** and **??**, Unit U04), $X = U\Sigma V^T$ gives

$$XX^T = U\Sigma\Sigma^T U^T = \sum_{i=1}^{r} \sigma_i^2 \boldsymbol{u}_i \boldsymbol{u}_i^T$$

where $\lambda_i = \sigma_i^2 > 0$ are the positive eigenvalues of XX^T

Conversely, *any* symmetric PSD matrix B can be factored as a Gram matrix $B = XX^T$ by reversing this step, as shown by Theorem 99 later

Before stating that theorem, we need to define the square root of a matrix...

Square root of a matrix

A square matrix S is called a **square root** of A if $S^2 = SS = A$, and we write $S = A^{1/2}$

For a diagonal matrix with non-negative diagonal entries $\lambda_i \geq 0$, its square root is obtained by simply taking the square root of each diagonal entry:

$$\Lambda = \text{diag}(\lambda_1, \ldots, \lambda_n) \implies \Lambda^{1/2} = \text{diag}(\sqrt{\lambda_1}, \ldots, \sqrt{\lambda_n})$$

Because a symmetric PSD matrix has eigenvalues $\lambda_i \geq 0$, the square roots $\sqrt{\lambda_i}$ are always real numbers

Example (a simple 2×2 diagonal matrix)

$$\Lambda = \begin{pmatrix} 4 & 0 \\ 0 & 9 \end{pmatrix} \implies \Lambda^{1/2} = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$$

since

$$\begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix} = \begin{pmatrix} 2^2 & 0 \\ 0 & 3^2 \end{pmatrix} = \begin{pmatrix} 4 & 0 \\ 0 & 9 \end{pmatrix}$$

Theorem 99 (Spectral factorisation of symmetric PSD matrices)

Let $B \in \mathcal{M}_n(\mathbb{R})$ be symmetric and positive semi-definite with $\operatorname{rank}(B) = r \leq n$. *Let its ordered eigenvalues be*

$$\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_r > 0 \quad \textit{and} \quad \lambda_{r+1} = \cdots = \lambda_n = 0$$

with corresponding orthonormal eigenvectors $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n \in \mathbb{R}^n$. Define

$$V_r = \begin{pmatrix} \boldsymbol{v}_1 & \cdots & \boldsymbol{v}_r \end{pmatrix} \in \mathcal{M}_{nr}(\mathbb{R}), \qquad \Lambda_r = \operatorname{diag}(\lambda_1, \ldots, \lambda_r) \in \mathcal{M}_r(\mathbb{R})$$

Then the matrix $X = V_r \Lambda_r^{1/2} \in \mathcal{M}_{nr}(\mathbb{R})$ satisfies $XX^T = B$

Hey, lookee here, another matrix factorisation :) Conditions are much more restrictive than in past factorisations, though, since we need a PSD matrix as input

Gram matrices and Euclidean distance matrices

The Gram matrix

The squared Euclidean distance matrix

Euclidean and squared Euclidean distance matrices

Definition 100 (Euclidean distance matrix and squared distance matrix)

Let $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n \in \mathbb{R}^p$ be the rows of $X \in \mathcal{M}_{np}(\mathbb{R})$

- *The **Euclidean distance matrix** is $D = [d_{ij}] \in \mathcal{M}_n(\mathbb{R})$, where*

$$d_{ij} = \|\boldsymbol{x}_i - \boldsymbol{x}_j\| = \sqrt{\sum_{m=1}^{p} (x_{im} - x_{jm})^2}, \qquad i, j = 1, \ldots, n$$

- *The **squared Euclidean distance matrix** is $D^{(2)} = [d_{ij}^2] \in \mathcal{M}_n(\mathbb{R})$, obtained by squaring each entry of D elementwise (**Hadamard square** $D^{(2)} = D \circ D$):*

$$\left[D^{(2)}\right]_{ij} = d_{ij}^2 = \|\boldsymbol{x}_i - \boldsymbol{x}_j\|^2, \qquad i, j = 1, \ldots, n$$

Warning: $D^{(2)}$ denotes the elementwise square $[d_{ij}^2]$, *not* the matrix product DD!

Basic properties of distance matrices

For any configuration of points $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n \in \mathbb{R}^p$, the matrices $D = [d_{ij}]$ and $D^{(2)} = [d_{ij}^2]$ satisfy:

1. **Zero diagonal** (**hollow matrix**):

$$d_{ii} = \|\boldsymbol{x}_i - \boldsymbol{x}_i\| = 0 \implies \operatorname{diag}(D) = \operatorname{diag}(D^{(2)}) = \mathbf{0} \quad \text{and} \quad \operatorname{tr}(D) = \operatorname{tr}(D^{(2)}) = 0$$

2. **Symmetry**:

$$d_{ij} = \|\boldsymbol{x}_i - \boldsymbol{x}_j\| = \|\boldsymbol{x}_j - \boldsymbol{x}_i\| = d_{ji} \implies D^T = D \quad \text{and} \quad (D^{(2)})^T = D^{(2)}$$

3. **Non-negativity**: $d_{ij} \geq 0$ and $d_{ij}^2 \geq 0$ for all $i, j = 1, \ldots, n$, with $d_{ij} = 0 \iff \boldsymbol{x}_i = \boldsymbol{x}_j$
4. **Triangle inequality** (for D, though *not* necessarily for $D^{(2)}$):

$$d_{ij} \leq d_{ik} + d_{kj} \qquad \forall i, j, k = 1, \ldots, n$$

Running example: squared distance matrix of the square

We previously found the Euclidean distance matrix D of our running example. Squaring each entry elementwise (d_{ij}^2) gives the **squared Euclidean distance matrix** $D^{(2)} \in \mathcal{M}_4(\mathbb{R})$:

$$D^{(2)} = D \circ D = \begin{pmatrix} 0^2 & 1^2 & (\sqrt{2})^2 & 1^2 \\ 1^2 & 0^2 & 1^2 & (\sqrt{2})^2 \\ (\sqrt{2})^2 & 1^2 & 0^2 & 1^2 \\ 1^2 & (\sqrt{2})^2 & 1^2 & 0^2 \end{pmatrix} = \begin{pmatrix} 0 & 1 & 2 & 1 \\ 1 & 0 & 1 & 2 \\ 2 & 1 & 0 & 1 \\ 1 & 2 & 1 & 0 \end{pmatrix}$$

As expected from the basic properties:

- $D^{(2)}$ is symmetric: $(D^{(2)})^T = D^{(2)}$
- $D^{(2)}$ has a zero diagonal: $\mathrm{diag}(D^{(2)}) = \mathbf{0}$ and $\mathrm{tr}(D^{(2)}) = 0$
- All off-diagonal entries are strictly positive

Relating squared distances to inner products

Why do we work with the *squared* distance matrix $D^{(2)}$ rather than D? Because squared Euclidean norms expand directly in terms of inner products!

For any two points $\boldsymbol{x}_i, \boldsymbol{x}_j \in \mathbb{R}^p$, using the bilinearity and symmetry of the inner product:

$$\begin{aligned} d_{ij}^2 = \|\boldsymbol{x}_i - \boldsymbol{x}_j\|^2 &= (\boldsymbol{x}_i - \boldsymbol{x}_j) \bullet (\boldsymbol{x}_i - \boldsymbol{x}_j) \\ &= (\boldsymbol{x}_i - \boldsymbol{x}_j)^T (\boldsymbol{x}_i - \boldsymbol{x}_j) \\ &= \boldsymbol{x}_i^T \boldsymbol{x}_i - \boldsymbol{x}_i^T \boldsymbol{x}_j - \boldsymbol{x}_j^T \boldsymbol{x}_i + \boldsymbol{x}_j^T \boldsymbol{x}_j \\ &= \|\boldsymbol{x}_i\|^2 - 2\boldsymbol{x}_i^T \boldsymbol{x}_j + \|\boldsymbol{x}_j\|^2 \end{aligned}$$

In terms of the entries $g_{ij} = \boldsymbol{x}_i^T \boldsymbol{x}_j$ of the Gram matrix $G = XX^T$:

$$d_{ij}^2 = g_{ii} - 2g_{ij} + g_{jj} \qquad \forall i, j = 1, \ldots, n$$

Notice that this is the law of cosines: $c^2 = a^2 + b^2 - 2ab\cos\theta$

Matrix formula for $D^{(2)}$ in terms of G

We can write the entrywise identity $d_{ij}^2 = g_{ii} - 2g_{ij} + g_{jj}$ as a single matrix equation

Theorem 101 (Squared distance matrix from the Gram matrix)

Let $X \in \mathcal{M}_{np}(\mathbb{R})$ have Gram matrix $G = XX^T \in \mathcal{M}_n(\mathbb{R})$. Let

$$\boldsymbol{g} = \operatorname{diag}(G) = \begin{pmatrix} g_{11} \\ g_{22} \\ \vdots \\ g_{nn} \end{pmatrix} = \begin{pmatrix} \|\boldsymbol{x}_1\|^2 \\ \|\boldsymbol{x}_2\|^2 \\ \vdots \\ \|\boldsymbol{x}_n\|^2 \end{pmatrix} \in \mathbb{R}^n \quad \textit{and} \quad \mathbb{1} = \begin{pmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix} \in \mathbb{R}^n$$

Then the squared Euclidean distance matrix $D^{(2)} = [d_{ij}^2] \in \mathcal{M}_n(\mathbb{R})$ is given by

$$D^{(2)} = \boldsymbol{g}\mathbb{1}^T - 2G + \mathbb{1}\boldsymbol{g}^T$$

Proof of Theorem 101

Examine the (i,j) entry of the rank-one outer product matrices $\boldsymbol{g}\mathbb{1}^T$ and $\mathbb{1}\boldsymbol{g}^T$:

$$\boldsymbol{g}\mathbb{1}^T = \begin{pmatrix} g_{11} \\ g_{22} \\ \vdots \\ g_{nn} \end{pmatrix} \begin{pmatrix} 1 & 1 & \cdots & 1 \end{pmatrix} = \begin{pmatrix} g_{11} & g_{11} & \cdots & g_{11} \\ g_{22} & g_{22} & \cdots & g_{22} \\ \vdots & \vdots & \ddots & \vdots \\ g_{nn} & g_{nn} & \cdots & g_{nn} \end{pmatrix} \implies \left[\boldsymbol{g}\mathbb{1}^T\right]_{ij} = g_{ii}$$

Similarly, taking the transpose:

$$\mathbb{1}\boldsymbol{g}^T = (\boldsymbol{g}\mathbb{1}^T)^T = \begin{pmatrix} g_{11} & g_{22} & \cdots & g_{nn} \\ g_{11} & g_{22} & \cdots & g_{nn} \\ \vdots & \vdots & \ddots & \vdots \\ g_{11} & g_{22} & \cdots & g_{nn} \end{pmatrix} \implies \left[\mathbb{1}\boldsymbol{g}^T\right]_{ij} = g_{jj}$$

Therefore, for every $(i,j) \in \{1,\ldots,n\}^2$,

$$\left[\boldsymbol{g}\mathbb{1}^T - 2G + \mathbb{1}\boldsymbol{g}^T\right]_{ij} = g_{ii} - 2g_{ij} + g_{jj} = \|\boldsymbol{x}_i\|^2 - 2\boldsymbol{x}_i^T\boldsymbol{x}_j + \|\boldsymbol{x}_j\|^2 = \|\boldsymbol{x}_i - \boldsymbol{x}_j\|^2 = d_{ij}^2 \quad \square$$

Running example: computing $D^{(2)}$ from G

We have already computed the Gram matrix $G = XX^T$ of our running example:

$$G = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & 1 \end{pmatrix} \implies \boldsymbol{g} = \operatorname{diag}(G) = \begin{pmatrix} 0 \\ 1 \\ 2 \\ 1 \end{pmatrix}$$

Using Theorem 101, we form $\boldsymbol{g}\mathbb{1}^T$, $\mathbb{1}\boldsymbol{g}^T$ and compute $D^{(2)} = \boldsymbol{g}\mathbb{1}^T - 2G + \mathbb{1}\boldsymbol{g}^T$:

$$D^{(2)} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 2 & 2 & 2 & 2 \\ 1 & 1 & 1 & 1 \end{pmatrix} - 2 \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & 1 \end{pmatrix} + \begin{pmatrix} 0 & 1 & 2 & 1 \\ 0 & 1 & 2 & 1 \\ 0 & 1 & 2 & 1 \\ 0 & 1 & 2 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 1 & 2 & 1 \\ 1 & 0 & 1 & 2 \\ 2 & 1 & 0 & 1 \\ 1 & 2 & 1 & 0 \end{pmatrix}$$

This matches the squared Euclidean distance matrix $D^{(2)}$ of the square we found earlier!

Example – Computing G and $D^{(2)}$ from X (1/2)

Consider $n = 3$ points in $\mathbb{R}^2$ given by the rows of

$$X = \begin{pmatrix} 1 & 2 \\ 3 & 1 \\ -1 & 0 \end{pmatrix} \in \mathcal{M}_{32}(\mathbb{R}), \qquad \text{i.e., } \boldsymbol{x}_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \boldsymbol{x}_2 = \begin{pmatrix} 3 \\ 1 \end{pmatrix}, \boldsymbol{x}_3 = \begin{pmatrix} -1 \\ 0 \end{pmatrix}$$

First, compute the Gram matrix $G = XX^T$:

$$G = \begin{pmatrix} 1 & 2 \\ 3 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 3 & -1 \\ 2 & 1 & 0 \end{pmatrix} = \begin{pmatrix} 5 & 5 & -1 \\ 5 & 10 & -3 \\ -1 & -3 & 1 \end{pmatrix}$$

The vector of diagonal entries of G (squared norms of the points) is

$$\boldsymbol{g} = \operatorname{diag}(G) = \begin{pmatrix} 5 \\ 10 \\ 1 \end{pmatrix}$$

Example – Computing G and $D^{(2)}$ from X (2/2)

Using Theorem 101, we form $\boldsymbol{g}\mathbb{1}^T$, $\mathbb{1}\boldsymbol{g}^T$ and $D^{(2)} = \boldsymbol{g}\mathbb{1}^T - 2G + \mathbb{1}\boldsymbol{g}^T$:

$$\boldsymbol{g}\mathbb{1}^T = \begin{pmatrix} 5 & 5 & 5 \\ 10 & 10 & 10 \\ 1 & 1 & 1 \end{pmatrix}, \qquad \mathbb{1}\boldsymbol{g}^T = \begin{pmatrix} 5 & 10 & 1 \\ 5 & 10 & 1 \\ 5 & 10 & 1 \end{pmatrix}$$

Hence:

$$D^{(2)} = \begin{pmatrix} 5 & 5 & 5 \\ 10 & 10 & 10 \\ 1 & 1 & 1 \end{pmatrix} - 2\begin{pmatrix} 5 & 5 & -1 \\ 5 & 10 & -3 \\ -1 & -3 & 1 \end{pmatrix} + \begin{pmatrix} 5 & 10 & 1 \\ 5 & 10 & 1 \\ 5 & 10 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 5 & 8 \\ 5 & 0 & 17 \\ 8 & 17 & 0 \end{pmatrix}$$

Check directly:

- $d_{12}^2 = \|(1,2)^T - (3,1)^T\|^2 = (-2)^2 + 1^2 = 5$
- $d_{13}^2 = \|(1,2)^T - (-1,0)^T\|^2 = 2^2 + 2^2 = 8$
- $d_{23}^2 = \|(3,1)^T - (-1,0)^T\|^2 = 4^2 + 1^2 = 17$

Motivation: from distances back to coordinates

Gram matrices and Euclidean distance matrices

The centring matrix as an orthogonal projector

Classical multidimensional scaling (MDS)

Algebraic duality between PCA and classical MDS

The all-ones vector $\mathbb{1}$ and the all-ones matrix $\mathbb{J}_n$

Two simple building blocks in $\mathbb{R}^n$ and $\mathcal{M}_n(\mathbb{R})$:

- The **all-ones vector** $\mathbb{1} \in \mathbb{R}^n$:

$$\mathbb{1} = \begin{pmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix} \in \mathbb{R}^n \implies \mathbb{1}^T\mathbb{1} = \sum_{i=1}^{n} 1^2 = n \quad \text{and} \quad \|\mathbb{1}\| = \sqrt{n}$$

- The **all-ones matrix** $\mathbb{J}_n = \mathbb{1}\mathbb{1}^T \in \mathcal{M}_n(\mathbb{R})$:

$$\mathbb{J}_n = \begin{pmatrix} 1 & 1 & \cdots & 1 \\ 1 & 1 & \cdots & 1 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & \cdots & 1 \end{pmatrix} \in \mathcal{M}_n(\mathbb{R}) \implies \mathbb{J}_n^2 = (\mathbb{1}\mathbb{1}^T)(\mathbb{1}\mathbb{1}^T) = \mathbb{1}(\mathbb{1}^T\mathbb{1})\mathbb{1}^T = n\mathbb{J}_n$$

Definition of the centring matrix

Definition 102 (Centring matrix)

*For $n \geq 1$, the **centring matrix** of order n is the $n \times n$ matrix*

$$C_n = \mathbb{I}_n - \frac{1}{n}\mathbb{1}\mathbb{1}^T = \mathbb{I}_n - \frac{1}{n}\mathbb{J}_n \in \mathcal{M}_n(\mathbb{R})$$

where $\mathbb{I}_n$ is the $n \times n$ identity matrix, $\mathbb{1} = (1, \ldots, 1)^T \in \mathbb{R}^n$ and $\mathbb{J}_n = \mathbb{1}\mathbb{1}^T$ is the $n \times n$ matrix of all ones

Entrywise, the (i, j) entry of C_n is

$$[C_n]_{ij} = \delta_{ij} - \frac{1}{n} = \begin{cases} 1 - \dfrac{1}{n} = \dfrac{n-1}{n} & \text{if } i = j \\ -\dfrac{1}{n} & \text{if } i \neq j \end{cases}$$

Explicit examples of C_n for small n

Writing out $C_n = \mathbb{I}_n - \frac{1}{n}\mathbb{J}_n$ explicitly for $n = 2, 3, 4$:

$$C_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} - \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} = \frac{1}{2}\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$$

$$C_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} - \frac{1}{3}\begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix} = \frac{1}{3}\begin{pmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{pmatrix}$$

$$C_4 = \frac{1}{4}\begin{pmatrix} 3 & -1 & -1 & -1 \\ -1 & 3 & -1 & -1 \\ -1 & -1 & 3 & -1 \\ -1 & -1 & -1 & 3 \end{pmatrix}$$

Notice that in every C_n, each row and each column sums to zero!

Algebraic and geometric properties of the centring matrix

Theorem 103 (Properties of the centring matrix C_n)

Let $C_n = \mathbb{I}_n - \frac{1}{n}\mathbb{1}\mathbb{1}^T \in \mathcal{M}_n(\mathbb{R})$. *Then:*

1. ***Symmetry****:* $C_n^T = C_n$
2. ***Idempotence****:* $C_n^2 = C_n$ *(and more generally* $C_n^k = C_n$ *for all integers* $k \geq 1$*)*
3. ***Null space****:* $C_n\mathbb{1} = \mathbf{0}$ *and* $\mathbb{1}^T C_n = \mathbf{0}^T$*; in fact,* $\text{null}(C_n) = \text{span}(\mathbb{1})$
4. ***Orthogonal projector****:* C_n *is the orthogonal projection matrix onto the* $(n-1)$*-dimensional subspace of zero-mean vectors*

$$W = \text{span}(\mathbb{1})^\perp = \left\{ \boldsymbol{v} \in \mathbb{R}^n : \mathbb{1}^T \boldsymbol{v} = \sum_{i=1}^{n} v_i = 0 \right\}$$

5. ***Spectrum, trace and rank****: the eigenvalues of* C_n *are* $\lambda = 0$ *(with multiplicity* 1*, eigenspace* $\text{span}(\mathbb{1})$*) and* $\lambda = 1$ *(with multiplicity* $n-1$*, eigenspace* W*). Consequently,* $\text{rank}(C_n) = \text{tr}(C_n) = n - 1$

Proof of Theorem 103 – Parts 1 and 2

1. C_n **is symmetric (**$C_n^T = C_n$**)**: Using linearity of the transpose, $\mathbb{I}_n^T = \mathbb{I}_n$ and $(\mathbb{1}\mathbb{1}^T)^T = (\mathbb{1}^T)^T\mathbb{1}^T = \mathbb{1}\mathbb{1}^T$:

$$C_n^T = \left(\mathbb{I}_n - \frac{1}{n}\mathbb{1}\mathbb{1}^T\right)^T = \mathbb{I}_n^T - \frac{1}{n}(\mathbb{1}\mathbb{1}^T)^T = \mathbb{I}_n - \frac{1}{n}\mathbb{1}\mathbb{1}^T = C_n \quad \square$$

2. C_n **is idempotent (**$C_n^2 = C_n$**)**: Expanding the square and using the associativity of matrix multiplication together with $\mathbb{1}^T\mathbb{1} = n$:

$$\begin{aligned} C_n^2 &= \left(\mathbb{I}_n - \frac{1}{n}\mathbb{1}\mathbb{1}^T\right)\left(\mathbb{I}_n - \frac{1}{n}\mathbb{1}\mathbb{1}^T\right) \\ &= \mathbb{I}_n^2 - \frac{1}{n}\mathbb{1}\mathbb{1}^T - \frac{1}{n}\mathbb{1}\mathbb{1}^T + \frac{1}{n^2}(\mathbb{1}\mathbb{1}^T)(\mathbb{1}\mathbb{1}^T) \\ &= \mathbb{I}_n - \frac{2}{n}\mathbb{1}\mathbb{1}^T + \frac{1}{n^2}\mathbb{1}(\underbrace{\mathbb{1}^T\mathbb{1}}_{=n})\mathbb{1}^T \\ &= \mathbb{I}_n - \frac{2}{n}\mathbb{1}\mathbb{1}^T + \frac{1}{n}\mathbb{1}\mathbb{1}^T = \mathbb{I}_n - \frac{1}{n}\mathbb{1}\mathbb{1}^T = C_n \quad \square \end{aligned}$$

Proof of Theorem 103 – Part 3

3. $C_n\mathbb{1} = \mathbf{0}$, $\mathbb{1}^T C_n = \mathbf{0}^T$ **and** $\text{null}(C_n) = \text{span}(\mathbb{1})$:
First, multiply C_n on the right by $\mathbb{1}$:

$$C_n\mathbb{1} = \left(\mathbb{I}_n - \frac{1}{n}\mathbb{1}\mathbb{1}^T\right)\mathbb{1} = \mathbb{1} - \frac{1}{n}\mathbb{1}(\mathbb{1}^T\mathbb{1}) = \mathbb{1} - \frac{1}{n}\mathbb{1}(n) = \mathbb{1} - \mathbb{1} = \mathbf{0}$$

Taking transposes and using $C_n^T = C_n$ immediately gives

$$\mathbb{1}^T C_n = \mathbb{1}^T C_n^T = (C_n\mathbb{1})^T = \mathbf{0}^T$$

To show $\text{null}(C_n) = \text{span}(\mathbb{1})$, let $\boldsymbol{v} \in \mathbb{R}^n$. Then

$$\boldsymbol{v} \in \text{null}(C_n) \iff C_n\boldsymbol{v} = \mathbf{0} \iff \boldsymbol{v} - \left(\frac{1}{n}\mathbb{1}^T\boldsymbol{v}\right)\mathbb{1} = \mathbf{0} \iff \boldsymbol{v} = \bar{v}\mathbb{1} \iff \boldsymbol{v} \in \text{span}(\mathbb{1})$$

Thus $\text{null}(C_n) = \text{span}(\mathbb{1})$, which is a 1-dimensional subspace of $\mathbb{R}^n$
(nullity$(C_n) = 1$) □

Proof of Theorem 103 – Part 4

4. C_n is the orthogonal projection matrix onto $W = \text{span}(\mathbb{1})^\perp$:
Let $L = \text{span}(\mathbb{1})$. Since $\{\mathbb{1}\}$ is an orthogonal basis of the 1-dimensional subspace L with $\|\mathbb{1}\|^2 = \mathbb{1}^T\mathbb{1} = n$, the orthogonal projection of any vector $\boldsymbol{v} \in \mathbb{R}^n$ onto L is

$$\text{proj}_L(\boldsymbol{v}) = \frac{\mathbb{1} \bullet \boldsymbol{v}}{\|\mathbb{1}\|^2}\mathbb{1} = \frac{\mathbb{1}^T\boldsymbol{v}}{n}\mathbb{1} = \left(\frac{1}{n}\mathbb{1}\mathbb{1}^T\right)\boldsymbol{v}$$

By definition of the orthogonal complement $W = L^\perp = \{\boldsymbol{v} \in \mathbb{R}^n : \mathbb{1}^T\boldsymbol{v} = 0\}$, the orthogonal projection of $\boldsymbol{v}$ onto W is the component of $\boldsymbol{v}$ orthogonal to L:

$$\text{proj}_W(\boldsymbol{v}) = \text{perp}_L(\boldsymbol{v}) = \boldsymbol{v} - \text{proj}_L(\boldsymbol{v}) = \boldsymbol{v} - \left(\frac{1}{n}\mathbb{1}\mathbb{1}^T\right)\boldsymbol{v} = \left(\mathbb{I}_n - \frac{1}{n}\mathbb{1}\mathbb{1}^T\right)\boldsymbol{v} = C_n\boldsymbol{v}$$

Note also that for any $\boldsymbol{v} \in \mathbb{R}^n$, the centred vector $C_n\boldsymbol{v}$ and the mean vector $\bar{v}\mathbb{1} = (\mathbb{I}_n - C_n)\boldsymbol{v}$ are orthogonal:

$$(C_n\boldsymbol{v}) \bullet (\bar{v}\mathbb{1}) = \bar{v}\boldsymbol{v}^T C_n^T\mathbb{1} = \bar{v}\boldsymbol{v}^T(C_n\mathbb{1}) = 0 \quad \square$$

Proof of Theorem 103 – Part 5

5. Eigenvalues, trace and rank of C_n:

Let $(\lambda, \boldsymbol{v})$ be an eigenpair of C_n ($\boldsymbol{v} \neq \mathbf{0}$ and $C_n\boldsymbol{v} = \lambda\boldsymbol{v}$). Applying C_n to both sides and using idempotence $C_n^2 = C_n$:

$$C_n\boldsymbol{v} = C_n^2\boldsymbol{v} = C_n(C_n\boldsymbol{v}) = C_n(\lambda\boldsymbol{v}) = \lambda(C_n\boldsymbol{v}) = \lambda^2\boldsymbol{v}$$

Since $\boldsymbol{v} \neq \mathbf{0}$, $(\lambda^2 - \lambda)\boldsymbol{v} = \mathbf{0} \implies \lambda(\lambda - 1) = 0$, so every eigenvalue of C_n is either $\lambda = 0$ or $\lambda = 1$:

- For $\lambda = 0$: the eigenspace is $\text{null}(C_n) = \text{span}(\mathbb{1})$, which has dimension 1 (multiplicity 1)
- For $\lambda = 1$: $\boldsymbol{v} \in W = \text{span}(\mathbb{1})^\perp \iff \mathbb{1}^T\boldsymbol{v} = 0 \implies C_n\boldsymbol{v} = \boldsymbol{v} - \frac{1}{n}\mathbb{1}(\mathbb{1}^T\boldsymbol{v}) = \boldsymbol{v}$, so the eigenspace for $\lambda = 1$ is W, which has dimension $n - 1$ (multiplicity $n - 1$)

Finally, the trace equals the sum of the diagonal entries (and the sum of the eigenvalues):

$$\text{tr}(C_n) = \sum_{i=1}^{n}\left(1 - \frac{1}{n}\right) = n\left(\frac{n-1}{n}\right) = n - 1 = \text{rank}(C_n) \quad \square$$

Action of C_n on a vector: subtracting the mean

Why is C_n called the *centring matrix*? Let $\boldsymbol{v} = (v_1, \ldots, v_n)^T \in \mathbb{R}^n$ be a column vector of n observations of a feature

First note that the inner product $\mathbb{1}^T \boldsymbol{v}$ is the sum of the entries of $\boldsymbol{v}$:

$$\mathbb{1}^T \boldsymbol{v} = \sum_{i=1}^{n} v_i = n\bar{v}, \qquad \text{where} \quad \bar{v} = \frac{1}{n} \sum_{i=1}^{n} v_i$$

Multiplying $\boldsymbol{v}$ on the left by C_n gives:

$$C_n \boldsymbol{v} = \left(\mathbb{I}_n - \frac{1}{n} \mathbb{1}\mathbb{1}^T \right) \boldsymbol{v} = \boldsymbol{v} - \frac{1}{n} \mathbb{1}(\mathbb{1}^T \boldsymbol{v}) = \boldsymbol{v} - \bar{v}\mathbb{1} = \begin{pmatrix} v_1 - \bar{v} \\ v_2 - \bar{v} \\ \vdots \\ v_n - \bar{v} \end{pmatrix}$$

Thus, $C_n \boldsymbol{v}$ subtracts the arithmetic mean $\bar{v}$ from every component of $\boldsymbol{v}$, producing a **zero-mean** (centred) vector

Action of C_n on a data matrix

Now let $X \in \mathcal{M}_{np}(\mathbb{R})$ be a data matrix with columns $\boldsymbol{c}_1, \ldots, \boldsymbol{c}_p \in \mathbb{R}^n$ (the p features) and rows $\boldsymbol{x}_1^T, \ldots, \boldsymbol{x}_n^T \in \mathbb{R}^p$ (the n observations)

- **Column-wise view**: left-multiplying X by C_n centres every column of X simultaneously:

$$\widetilde{X} = C_n X = \begin{pmatrix} C_n \boldsymbol{c}_1 & C_n \boldsymbol{c}_2 & \cdots & C_n \boldsymbol{c}_p \end{pmatrix}$$

- **Row-wise view**: if $\bar{\boldsymbol{x}} = \frac{1}{n} \sum_{i=1}^n \boldsymbol{x}_i = \frac{1}{n} X^T \mathbb{1} \in \mathbb{R}^p$ is the centroid of the n points, then

$$\widetilde{X} = C_n X = X - \frac{1}{n} \mathbb{1}\mathbb{1}^T X = X - \mathbb{1}\bar{\boldsymbol{x}}^T = \begin{pmatrix} (\boldsymbol{x}_1 - \bar{\boldsymbol{x}})^T \\ \vdots \\ (\boldsymbol{x}_n - \bar{\boldsymbol{x}})^T \end{pmatrix}$$

shifts the centroid of the n points to the origin $\mathbf{0} \in \mathbb{R}^p$

Link to covariance (Unit 05)

The sample covariance matrix $S_X \in \mathcal{M}_p(\mathbb{R})$ of X can be written compactly as

$$
\begin{aligned}
S_X &= \frac{1}{n-1} \widetilde{X}^T \widetilde{X} \\
&= \frac{1}{n-1} (C_n X)^T (C_n X) \\
&= \frac{1}{n-1} X^T C_n^T C_n X \\
&= \frac{1}{n-1} X^T C_n C_n X && \text{as } C_n \text{ is symmetric} \\
&= \frac{1}{n-1} X^T C_n X && \text{as } C_n \text{ is idempotent}
\end{aligned}
$$

Motivation: from distances back to coordinates

Gram matrices and Euclidean distance matrices

The centring matrix as an orthogonal projector

Classical multidimensional scaling (MDS)

Algebraic duality between PCA and classical MDS

Classical multidimensional scaling (MDS)

- **Double centring: recovering the Gram matrix from distances**
- Coordinate recovery via spectral decomposition
- Examples

From $D^{(2)}$ back to G: why double centring works

Recall Theorem 101: given the Gram matrix $G = XX^T$, the squared distance matrix is

$$D^{(2)} = \boldsymbol{g}\mathbb{1}^T - 2G + \mathbb{1}\boldsymbol{g}^T$$

We now want to **invert** this relationship: given only $D^{(2)}$, how can we isolate G?

Look closely at the two nuisance terms $\boldsymbol{g}\mathbb{1}^T$ and $\mathbb{1}\boldsymbol{g}^T$:

- $\boldsymbol{g}\mathbb{1}^T$ has $\mathbb{1}^T$ on the **right**, and we know that $\mathbb{1}^T C_n = \mathbf{0}^T$!
- $\mathbb{1}\boldsymbol{g}^T$ has $\mathbb{1}$ on the **left**, and we know that $C_n\mathbb{1} = \mathbf{0}$!

Therefore, if we multiply $D^{(2)}$ on the **left** by C_n and on the **right** by C_n, both rank-one nuisance terms $\boldsymbol{g}\mathbb{1}^T$ and $\mathbb{1}\boldsymbol{g}^T$ are completely annihilated!

This operation $D^{(2)} \mapsto -\frac{1}{2} C_n D^{(2)} C_n$ is called **double centring**

The double centring theorem

Theorem 104 (Double centring formula)

Let $X \in \mathcal{M}_{np}(\mathbb{R})$ be a data matrix with squared Euclidean distance matrix $D^{(2)} = [d_{ij}^2] \in \mathcal{M}_n(\mathbb{R})$, and let $\widetilde{X} = C_n X$ be the column-centred data matrix with centred Gram matrix $B = \widetilde{X}\widetilde{X}^T = C_n X X^T C_n$. Then

$$B = -\frac{1}{2} C_n D^{(2)} C_n = \widetilde{X}\widetilde{X}^T$$

In particular, if the points in X are already centred at the origin ($C_n X = X$, so $\sum_{i=1}^n \boldsymbol{x}_i = \mathbf{0}$), then $B = -\frac{1}{2} C_n D^{(2)} C_n = G = XX^T$ recovers the Gram matrix of X

Proof of Theorem 104

By Theorem 101, $D^{(2)} = \boldsymbol{g}\mathbb{1}^T - 2G + \mathbb{1}\boldsymbol{g}^T$, where $G = XX^T$ and $\boldsymbol{g} = \text{diag}(G)$
Multiply $D^{(2)}$ on the left by C_n and on the right by C_n:

$$C_n D^{(2)} C_n = C_n(\boldsymbol{g}\mathbb{1}^T)C_n - 2C_nGC_n + C_n(\mathbb{1}\boldsymbol{g}^T)C_n$$

Using associativity and Part 3 of Theorem 103 ($C_n\mathbb{1} = \mathbf{0}$ and $\mathbb{1}^T C_n = \mathbf{0}^T$):

$$C_n(\boldsymbol{g}\mathbb{1}^T)C_n = (C_n\boldsymbol{g})(\underbrace{\mathbb{1}^T C_n}_{=\mathbf{0}^T}) = (C_n\boldsymbol{g})\mathbf{0}^T = \mathbf{0}_{n\times n}$$

$$C_n(\mathbb{1}\boldsymbol{g}^T)C_n = (\underbrace{C_n\mathbb{1}}_{=\mathbf{0}})(\boldsymbol{g}^T C_n) = \mathbf{0}(\boldsymbol{g}^T C_n) = \mathbf{0}_{n\times n}$$

Thus both outer-product terms vanish, leaving

$$C_n D^{(2)} C_n = -2C_nGC_n = -2C_n(XX^T)C_n^T = -2(C_nX)(C_nX)^T = -2\widetilde{X}\widetilde{X}^T$$

Dividing both sides by -2 gives $-\frac{1}{2}C_nD^{(2)}C_n = C_nGC_n = \widetilde{X}\widetilde{X}^T$. If $C_nX = X$, then $\widetilde{X} = X$ and $-\frac{1}{2}C_nD^{(2)}C_n = XX^T = G$ □

Scalar form of double centring

Why is $B = -\frac{1}{2} C_n D^{(2)} C_n$ called *double* centring?

- Left-multiplying $D^{(2)}$ by C_n subtracts the **column means** of $D^{(2)}$ from each column
- Right-multiplying by $C_n = C_n^T$ subtracts the **row means** from each row

Let $\bar{d}^2_{i\cdot} = \frac{1}{n}\sum_{j=1}^n d^2_{ij}$ (mean of row i), $\bar{d}^2_{\cdot j} = \frac{1}{n}\sum_{i=1}^n d^2_{ij}$ (mean of column j) and $\bar{d}^2_{\cdot\cdot} = \frac{1}{n^2}\sum_{i=1}^n\sum_{j=1}^n d^2_{ij}$ (grand mean of $D^{(2)}$). Expanding $B = -\frac{1}{2}\left(\mathbb{I}_n - \frac{1}{n}\mathbb{J}_n\right) D^{(2)} \left(\mathbb{I}_n - \frac{1}{n}\mathbb{J}_n\right)$ gives the entrywise formula:

$$b_{ij} = -\frac{1}{2}\left(d^2_{ij} - \bar{d}^2_{i\cdot} - \bar{d}^2_{\cdot j} + \bar{d}^2_{\cdot\cdot}\right), \qquad i, j = 1, \ldots, n$$

Consequently, every row sum and every column sum of B is zero: $B\mathbb{1} = \mathbf{0}$ and $\mathbb{1}^T B = \mathbf{0}^T$, so $\lambda = 0$ is always an eigenvalue of B with eigenvector $\mathbb{1}$!

Classical multidimensional scaling (MDS)

- Double centring: recovering the Gram matrix from distances
- **Coordinate recovery via spectral decomposition**
- Examples

The classical MDS algorithm

Definition 105 (Classical MDS algorithm)

Given an $n \times n$ distance or dissimilarity matrix $D = [d_{ij}] \in \mathcal{M}_n(\mathbb{R})$ and a target embedding dimension $k \in \{1, \dots, n-1\}$:

1. ***Square the distances**: form $D^{(2)} = [d_{ij}^2] \in \mathcal{M}_n(\mathbb{R})$*
2. ***Double-centre**: compute the inner product matrix $B = -\frac{1}{2} C_n D^{(2)} C_n$*
3. ***Spectral decomposition**: compute the eigenvalues $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n$ and orthonormal eigenvectors $\boldsymbol{v}_1, \boldsymbol{v}_2, \dots, \boldsymbol{v}_n \in \mathbb{R}^n$ of B*
4. ***Construct k-dimensional coordinates**: assuming $\lambda_1 \geq \cdots \geq \lambda_k > 0$, let $V_k = [\boldsymbol{v}_1 \ \cdots \ \boldsymbol{v}_k] \in \mathcal{M}_{nk}(\mathbb{R})$ and $\Lambda_k = \operatorname{diag}(\lambda_1, \dots, \lambda_k) \in \mathcal{M}_k(\mathbb{R})$. The k-dimensional MDS coordinate matrix is*

$$X_k = V_k \Lambda_k^{1/2} = \begin{pmatrix} \sqrt{\lambda_1}\, \boldsymbol{v}_1 & \sqrt{\lambda_2}\, \boldsymbol{v}_2 & \cdots & \sqrt{\lambda_k}\, \boldsymbol{v}_k \end{pmatrix} \in \mathcal{M}_{nk}(\mathbb{R})$$

Classical multidimensional scaling (MDS)

- Double centring: recovering the Gram matrix from distances
- Coordinate recovery via spectral decomposition
- **Examples**

Running example: reconstructing the square from $D^{(2)}$ (1/15)

We now apply the classical MDS algorithm to our running example

Input: the squared Euclidean distance matrix $D^{(2)} \in \mathcal{M}_4(\mathbb{R})$ computed earlier:

$$D^{(2)} = \begin{pmatrix} 0 & 1 & 2 & 1 \\ 1 & 0 & 1 & 2 \\ 2 & 1 & 0 & 1 \\ 1 & 2 & 1 & 0 \end{pmatrix}$$

Objective:

1. Double-centre $D^{(2)}$ to obtain the inner product matrix $B = -\frac{1}{2} C_4 D^{(2)} C_4$
2. Find the eigenvalues and orthonormal eigenvectors of B
3. Deduce the intrinsic embedding dimension $r = \text{rank}(B)$
4. Construct the centred coordinate matrix $X_r \in \mathcal{M}_{4r}(\mathbb{R})$
5. Verify that pairwise distances are preserved and compare with $\widetilde{X}$

Running example: double-centring $D^{(2)}$ (2/15)

Step 1: Compute the row, column, and grand means of $D^{(2)}$
Using the entrywise double-centring formula:

$$b_{ij} = -\frac{1}{2}\left(d_{ij}^2 - \bar{d}_{i\cdot}^2 - \bar{d}_{\cdot j}^2 + \bar{d}_{\cdot\cdot}^2\right)$$

- **Row means**: every row of $D^{(2)}$ has the same sum $0 + 1 + 2 + 1 = 4$:

$$\bar{d}_{i\cdot}^2 = \frac{1}{4}(0 + 1 + 2 + 1) = \frac{4}{4} = 1, \qquad i = 1, 2, 3, 4$$

- **Column means**: by symmetry of $D^{(2)}$,

$$\bar{d}_{\cdot j}^2 = 1, \qquad j = 1, 2, 3, 4$$

- **Grand mean**:

$$\bar{d}_{\cdot\cdot}^2 = \frac{1}{4}\sum_{i=1}^{4} \bar{d}_{i\cdot}^2 = \frac{1}{4}(1 + 1 + 1 + 1) = 1$$

Running example: the double-centred Gram matrix B (3/15)

Step 2: Form the matrix $B = -\frac{1}{2} C_4 D^{(2)} C_4$

Substituting the means $\bar{d}^2_{i\cdot} = \bar{d}^2_{\cdot j} = \bar{d}^2_{\cdot\cdot} = 1$ into the entrywise formula:

$$b_{ij} = -\frac{1}{2}\left(d^2_{ij} - 1 - 1 + 1\right) = -\frac{1}{2}\left(d^2_{ij} - 1\right) = \frac{1 - d^2_{ij}}{2}$$

This yields the double-centred matrix:

$$B = -\frac{1}{2} C_4 D^{(2)} C_4 = \frac{1}{2}\begin{pmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \\ -1 & 0 & 1 & 0 \\ 0 & -1 & 0 & 1 \end{pmatrix}$$

Properties of B:

- **Centring**: row and column sums are zero, $B\mathbb{1} = \mathbf{0}$ and $\mathbb{1}^T B = \mathbf{0}^T$
- **Trace**: $\mathrm{tr}(B) = \frac{1}{2}(1 + 1 + 1 + 1) = 2$
- **Exact match**: B is identical to $\widetilde{X}\widetilde{X}^T$, the Gram matrix of the centred square $\widetilde{X} = C_4 X$ computed earlier!

Running example: rank and intrinsic dimension of B (4/15)

Step 3: Analyze the rank and geometry of

$$B = \frac{1}{2}\begin{pmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \\ -1 & 0 & 1 & 0 \\ 0 & -1 & 0 & 1 \end{pmatrix}$$

- **Row relationships**:

$$\text{row}_3(B) = -\text{row}_1(B) \qquad \text{and} \qquad \text{row}_4(B) = -\text{row}_2(B)$$

- **Linearly independent rows**: rows 1 and 2 are orthogonal and linearly independent
- **Rank**: therefore,

$$r = \text{rank}(B) = 2$$

- **Intrinsic dimension**: By Schoenberg's theorem, the 4 points lie intrinsically in an $r = 2$ dimensional Euclidean space (a plane), with zero out-of-plane distortion!

Running example: eigenvalues of B (5/15)

To find the eigenvalues of B, notice that pairs $\{1,3\}$ and $\{2,4\}$ decouple:

Permuting rows and columns to ordering $(1,3,2,4)$ yields the block-diagonal matrix:

$$PBP^T = \frac{1}{2}\begin{pmatrix} 1 & -1 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & -1 & 1 \end{pmatrix} = \begin{pmatrix} M & \mathbf{0} \\ \mathbf{0} & M \end{pmatrix}, \quad \text{where } M = \frac{1}{2}\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$$

- Each 2×2 block M has $\mathrm{tr}(M) = 1$ and $\det(M) = 0$, giving eigenvalues 1 and 0:

$$\det(M - \lambda \mathbb{I}_2) = \lambda^2 - \lambda = \lambda(\lambda - 1) = 0 \implies \lambda \in \{1, 0\}$$

- Since PBP^T has two identical blocks M, the 4 ordered eigenvalues of B are:

$$\lambda_1 = 1, \qquad \lambda_2 = 1, \qquad \lambda_3 = 0, \qquad \lambda_4 = 0$$

Running example: linear system for the positive eigenvalues (6/15)

The positive eigenvalue is $\lambda = 1$ with algebraic multiplicity 2

To find the corresponding eigenvectors, we solve $(B - 1 \cdot \mathbb{I}_4)\boldsymbol{v} = \mathbf{0}$:

$$\frac{1}{2}\begin{pmatrix} -1 & 0 & -1 & 0 \\ 0 & -1 & 0 & -1 \\ -1 & 0 & -1 & 0 \\ 0 & -1 & 0 & -1 \end{pmatrix}\begin{pmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

Multiplying out yields the linear system:

$$\begin{cases} -\frac{1}{2}(v_1 + v_3) = 0 \\ -\frac{1}{2}(v_2 + v_4) = 0 \\ -\frac{1}{2}(v_1 + v_3) = 0 \\ -\frac{1}{2}(v_2 + v_4) = 0 \end{cases} \iff \begin{cases} v_3 = -v_1 \\ v_4 = -v_2 \end{cases}$$

Thus v_1 and v_2 are free variables, while v_3 and v_4 are completely determined

Running example: the 2-dimensional eigenspace $E_B(1)$ (7/15)

With free parameters $v_1 = a \in \mathbb{R}$ and $v_2 = b \in \mathbb{R}$, every eigenvector $\boldsymbol{v}$ for $\lambda = 1$ has the form:

$$\boldsymbol{v} = \begin{pmatrix} a \\ b \\ -a \\ -b \end{pmatrix} = a \begin{pmatrix} 1 \\ 0 \\ -1 \\ 0 \end{pmatrix} + b \begin{pmatrix} 0 \\ 1 \\ 0 \\ -1 \end{pmatrix}, \qquad (a, b) \neq (0, 0)$$

The eigenspace associated with $\lambda = 1$ is therefore two-dimensional:

$$E_B(1) = \text{span} \left\{ \boldsymbol{u}_1 = \begin{pmatrix} 1 \\ 0 \\ -1 \\ 0 \end{pmatrix}, \ \boldsymbol{u}_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \\ -1 \end{pmatrix} \right\}$$

Geometric interpretation:

- $\boldsymbol{u}_1$ separates opposite vertices 1 and 3 along the first coordinate direction
- $\boldsymbol{u}_2$ separates opposite vertices 2 and 4 along the second coordinate direction

Running example: orthonormal eigenvectors of B (8/15)

To construct the coordinate matrix, we choose an **orthonormal basis** $\{\boldsymbol{v}_1, \boldsymbol{v}_2\}$ for $E_B(1)$:

$$\boldsymbol{v}_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ -1 \\ 0 \end{pmatrix}, \qquad \boldsymbol{v}_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ 0 \\ -1 \end{pmatrix}$$

Verification of orthonormality:

- **Unit norm**:

$$\|\boldsymbol{v}_1\|^2 = \frac{1}{2}(1^2 + 0^2 + (-1)^2 + 0^2) = 1, \quad \|\boldsymbol{v}_2\|^2 = \frac{1}{2}(0^2 + 1^2 + 0^2 + (-1)^2) = 1$$

- **Orthogonality**:

$$\boldsymbol{v}_1^T \boldsymbol{v}_2 = \frac{1}{2}(1 \cdot 0 + 0 \cdot 1 + (-1) \cdot 0 + 0 \cdot (-1)) = 0$$

Thus $V_2 = [\boldsymbol{v}_1 \;\; \boldsymbol{v}_2] \in \mathcal{M}_{42}(\mathbb{R})$ has orthonormal columns: $V_2^T V_2 = \mathbb{I}_2$

Running example: coordinate recovery (9/15)

Step 4: With $k = 2$ and $\Lambda_2 = \text{diag}(1, 1) = \mathbb{I}_2$, the coordinate matrix is:

$$X_2 = V_2 \Lambda_2^{1/2} = \begin{pmatrix} \boldsymbol{v}_1 & \boldsymbol{v}_2 \end{pmatrix} \mathbb{I}_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ -1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} \frac{\sqrt{2}}{2} & 0 \\ 0 & \frac{\sqrt{2}}{2} \\ -\frac{\sqrt{2}}{2} & 0 \\ 0 & -\frac{\sqrt{2}}{2} \end{pmatrix} \in \mathcal{M}_{42}(\mathbb{R})$$

The reconstructed points in $\mathbb{R}^2$ (the rows of X_2) are:

$$\hat{\boldsymbol{x}}_1 = \begin{pmatrix} \frac{\sqrt{2}}{2} \\ 0 \end{pmatrix}, \qquad \hat{\boldsymbol{x}}_2 = \begin{pmatrix} 0 \\ \frac{\sqrt{2}}{2} \end{pmatrix}, \qquad \hat{\boldsymbol{x}}_3 = \begin{pmatrix} -\frac{\sqrt{2}}{2} \\ 0 \end{pmatrix}, \qquad \hat{\boldsymbol{x}}_4 = \begin{pmatrix} 0 \\ -\frac{\sqrt{2}}{2} \end{pmatrix}$$

Running example: verification of centroid and edges (10/15)

We verify that the reconstructed configuration X_2 satisfies all requirements:

- **Centroid at origin**:

$$\bar{\hat{\boldsymbol{x}}} = \frac{1}{4}\sum_{i=1}^{4} \hat{\boldsymbol{x}}_i = \frac{1}{4}\begin{pmatrix} \frac{\sqrt{2}}{2} + 0 - \frac{\sqrt{2}}{2} + 0 \\ 0 + \frac{\sqrt{2}}{2} + 0 - \frac{\sqrt{2}}{2} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

- **Adjacent vertices (edge lengths)**:

$$\|\hat{\boldsymbol{x}}_1 - \hat{\boldsymbol{x}}_2\|^2 = \left(\frac{\sqrt{2}}{2} - 0\right)^2 + \left(0 - \frac{\sqrt{2}}{2}\right)^2 = \frac{1}{2} + \frac{1}{2} = 1 \implies \hat{d}_{12} = 1$$

$$\|\hat{\boldsymbol{x}}_2 - \hat{\boldsymbol{x}}_3\|^2 = \left(0 - \left(-\frac{\sqrt{2}}{2}\right)\right)^2 + \left(\frac{\sqrt{2}}{2} - 0\right)^2 = \frac{1}{2} + \frac{1}{2} = 1 \implies \hat{d}_{23} = 1$$

By symmetry, $\hat{d}_{34} = 1$ and $\hat{d}_{41} = 1$. All four edge lengths are 1!

Running example: verification of diagonals and distance matrix (11/15)

- **Opposite vertices (diagonal lengths)**:

$$\|\hat{\boldsymbol{x}}_1 - \hat{\boldsymbol{x}}_3\|^2 = \left(\frac{\sqrt{2}}{2} - \left(-\frac{\sqrt{2}}{2}\right)\right)^2 + (0-0)^2 = (\sqrt{2})^2 = 2 \implies \hat{d}_{13} = \sqrt{2}$$

$$\|\hat{\boldsymbol{x}}_2 - \hat{\boldsymbol{x}}_4\|^2 = (0-0)^2 + \left(\frac{\sqrt{2}}{2} - \left(-\frac{\sqrt{2}}{2}\right)\right)^2 = (\sqrt{2})^2 = 2 \implies \hat{d}_{24} = \sqrt{2}$$

- **Reconstructed distance matrices**:

$$\widehat{D} = \begin{pmatrix} 0 & 1 & \sqrt{2} & 1 \\ 1 & 0 & 1 & \sqrt{2} \\ \sqrt{2} & 1 & 0 & 1 \\ 1 & \sqrt{2} & 1 & 0 \end{pmatrix} = D, \qquad \widehat{D}^{(2)} = \begin{pmatrix} 0 & 1 & 2 & 1 \\ 1 & 0 & 1 & 2 \\ 2 & 1 & 0 & 1 \\ 1 & 2 & 1 & 0 \end{pmatrix} = D^{(2)}$$

- Pairwise distances are **isometrically and perfectly reconstructed**!

Running example: algebraic comparison with centred square (12/15)

Why did classical MDS yield vertices on the coordinate axes rather than $[\pm 1/2, \pm 1/2]^2$?

- Recall the original square centred at the origin ($\widetilde{X} = C_4 X$):

$$\tilde{\boldsymbol{x}}_1 = \begin{pmatrix} -1/2 \\ -1/2 \end{pmatrix}, \quad \tilde{\boldsymbol{x}}_2 = \begin{pmatrix} 1/2 \\ -1/2 \end{pmatrix}, \quad \tilde{\boldsymbol{x}}_3 = \begin{pmatrix} 1/2 \\ 1/2 \end{pmatrix}, \quad \tilde{\boldsymbol{x}}_4 = \begin{pmatrix} -1/2 \\ 1/2 \end{pmatrix}$$

- Both configurations share the exact same distance from the origin:

$$\|\tilde{\boldsymbol{x}}_i\| = \sqrt{(-1/2)^2 + (-1/2)^2} = \frac{\sqrt{2}}{2} = \|\hat{\boldsymbol{x}}_i\|$$

- The reconstructed configuration X_2 is related to $\widetilde{X}$ by a rigid rotation by $\theta = -45^\circ$ $(-\pi/4)$:

$$Q = \begin{pmatrix} \cos(-\pi/4) & -\sin(-\pi/4) \\ \sin(-\pi/4) & \cos(-\pi/4) \end{pmatrix} = \frac{\sqrt{2}}{2} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} \in \mathcal{SO}(2)$$

Running example: explicit rotation calculation (13/15)

Applying the rotation matrix $Q = \frac{\sqrt{2}}{2}\begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$ to each centred point $\tilde{\boldsymbol{x}}_i$:

- Vertices 1 and 2:

$$Q\tilde{\boldsymbol{x}}_1 = \frac{\sqrt{2}}{2}\begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}\begin{pmatrix} -1/2 \\ -1/2 \end{pmatrix} = \begin{pmatrix} -\frac{\sqrt{2}}{2} \\ 0 \end{pmatrix} = \hat{\boldsymbol{x}}_3$$

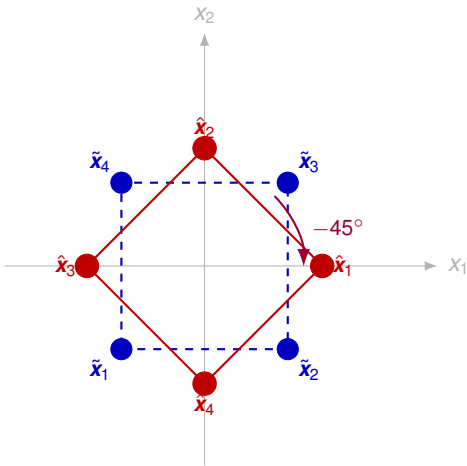
$$Q\tilde{\boldsymbol{x}}_2 = \frac{\sqrt{2}}{2}\begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}\begin{pmatrix} 1/2 \\ -1/2 \end{pmatrix} = \begin{pmatrix} 0 \\ -\frac{\sqrt{2}}{2} \end{pmatrix} = \hat{\boldsymbol{x}}_4$$

- Vertices 3 and 4:

$$Q\tilde{\boldsymbol{x}}_3 = \frac{\sqrt{2}}{2}\begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}\begin{pmatrix} 1/2 \\ 1/2 \end{pmatrix} = \begin{pmatrix} \frac{\sqrt{2}}{2} \\ 0 \end{pmatrix} = \hat{\boldsymbol{x}}_1$$

$$Q\tilde{\boldsymbol{x}}_4 = \frac{\sqrt{2}}{2}\begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}\begin{pmatrix} -1/2 \\ 1/2 \end{pmatrix} = \begin{pmatrix} 0 \\ \frac{\sqrt{2}}{2} \end{pmatrix} = \hat{\boldsymbol{x}}_2$$

Running example: geometric visualization (14/15)



Running example: fundamental takeaways of classical MDS (15/15)

Key insights illustrated by the running example:

- **Exact isometric reconstruction**: When D is an exact Euclidean distance matrix, classical MDS recovers the configuration with zero distortion in $r = \text{rank}(B)$ dimensions
- **Centroid at origin**: The reconstructed configuration is automatically centred: $\sum_{i=1}^{n} \hat{\boldsymbol{x}}_i = \mathbf{0}$
- **Orthogonal indeterminacy**: The reconstruction is unique only **up to an orthogonal transformation** (rotation or reflection about the origin)
- **Basis of the eigenspace**: Any orthonormal basis of $E_B(1)$ yields a valid MDS embedding; different choices of basis simply correspond to different orientations (rotations/reflections) of the same shape

Example 1 – Recovering 1D coordinates from $D^{(2)}$ (1/4)

Suppose $n = 3$ objects have the squared Euclidean distance matrix

$$D^{(2)} = \begin{pmatrix} 0 & 4 & 16 \\ 4 & 0 & 4 \\ 16 & 4 & 0 \end{pmatrix}$$

(i.e. $d_{12} = 2$, $d_{23} = 2$ and $d_{13} = 4$)

Problem:

1. Compute the double-centred Gram matrix $B = -\frac{1}{2} C_3 D^{(2)} C_3$
2. Find the eigenvalues and orthonormal eigenvectors of B
3. Determine the minimal embedding dimension $r = \text{rank}(B)$ and recover the centred coordinates X_r

Example 1 – Computing $B = -\frac{1}{2} C_3 D^{(2)} C_3$ (2/4)

Recall $C_3 = \frac{1}{3}\begin{pmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{pmatrix}$. First compute $C_3 D^{(2)}$:

$$C_3 D^{(2)} = \frac{1}{3}\begin{pmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{pmatrix}\begin{pmatrix} 0 & 4 & 16 \\ 4 & 0 & 4 \\ 16 & 4 & 0 \end{pmatrix} = \frac{1}{3}\begin{pmatrix} -20 & 4 & 28 \\ -16 & -8 & -16 \\ 36 & 4 & -12 \end{pmatrix} = \frac{4}{3}\begin{pmatrix} -5 & 1 & 7 \\ -4 & -2 & -4 \\ 9 & 1 & -3 \end{pmatrix}$$

Now right-multiply by C_3 and multiply by $-\frac{1}{2}$:

$$C_3 D^{(2)} C_3 = \frac{4}{9}\begin{pmatrix} -5 & 1 & 7 \\ -4 & -2 & -4 \\ 9 & 1 & -3 \end{pmatrix}\begin{pmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{pmatrix} = \frac{4}{9}\begin{pmatrix} -18 & 0 & 18 \\ 0 & 0 & 0 \\ 18 & 0 & -18 \end{pmatrix} = \begin{pmatrix} -8 & 0 & 8 \\ 0 & 0 & 0 \\ 8 & 0 & -8 \end{pmatrix}$$

$$\implies B = -\frac{1}{2} C_3 D^{(2)} C_3 = \begin{pmatrix} 4 & 0 & -4 \\ 0 & 0 & 0 \\ -4 & 0 & 4 \end{pmatrix}$$

Example 1 – Eigenpairs and coordinate recovery (3/4)

We have

$$B = \begin{pmatrix} 4 & 0 & -4 \\ 0 & 0 & 0 \\ -4 & 0 & 4 \end{pmatrix}$$

- Notice that row 1 and row 3 are opposites and row 2 is zero, so $\text{rank}(B) = 1$
- Since $\text{tr}(B) = 4 + 0 + 4 = 8$ and $\text{rank}(B) = 1$, the eigenvalues of B are immediately

$$\lambda_1 = 8, \qquad \lambda_2 = 0, \qquad \lambda_3 = 0$$

- For $\lambda_1 = 8$, $(B - 8\mathbb{I}_3)\boldsymbol{v}_1 = \mathbf{0} \implies v_{11} + v_{13} = 0$ and $v_{12} = 0$. Normalising gives the unit eigenvector

$$\boldsymbol{v}_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} \qquad (\text{or } -\boldsymbol{v}_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix})$$

Example 1 – Verification of recovered 1D coordinates (4/4)

Since B is positive semi-definite with $r = \text{rank}(B) = 1$, the 3 points lie on a 1-dimensional line ($\mathbb{R}^1$)!

Recovering the 1-dimensional coordinate vector $X_1 = \sqrt{\lambda_1}\, \boldsymbol{v}_1 \in \mathcal{M}_{31}(\mathbb{R})$:

$$X_1 = \sqrt{8}\left(\frac{1}{\sqrt{2}}\begin{pmatrix}-1\\0\\1\end{pmatrix}\right) = \frac{2\sqrt{2}}{\sqrt{2}}\begin{pmatrix}-1\\0\\1\end{pmatrix} = \begin{pmatrix}-2\\0\\2\end{pmatrix}$$

Check:

- Centred at the origin: $x_1 + x_2 + x_3 = -2 + 0 + 2 = 0$
- Pairwise squared distances:

$$(x_1 - x_2)^2 = (-2-0)^2 = 4, \quad (x_2 - x_3)^2 = (0-2)^2 = 4, \quad (x_1 - x_3)^2 = (-2-2)^2 = 16$$

which reproduces $D^{(2)}$ exactly! (Choosing $-\boldsymbol{v}_1$ gives $X_1 = (2, 0, -2)^T$, the reflection across the origin)

Example 2 – Recovering a 2D triangle from distances (1/5)

Now consider $n = 3$ points with squared Euclidean distance matrix

$$D^{(2)} = \begin{pmatrix} 0 & 6 & 6 \\ 6 & 0 & 12 \\ 6 & 12 & 0 \end{pmatrix}$$

(an isosceles right triangle with leg length $\sqrt{6}$ and hypotenuse $\sqrt{12}$, since $6 + 6 = 12$)

Let us compute $B = -\frac{1}{2} C_3 D^{(2)} C_3$ using the row/column mean formula $b_{ij} = -\frac{1}{2}(d_{ij}^2 - \bar{d}_{i\cdot}^2 - \bar{d}_{\cdot j}^2 + \bar{d}_{\cdot\cdot}^2)$:

- Row (and column) means of $D^{(2)}$:

$$\bar{d}_{1\cdot}^2 = \frac{0 + 6 + 6}{3} = 4, \qquad \bar{d}_{2\cdot}^2 = \frac{6 + 0 + 12}{3} = 6, \qquad \bar{d}_{3\cdot}^2 = \frac{6 + 12 + 0}{3} = 6$$

- Grand mean of $D^{(2)}$:

$$\bar{d}_{\cdot\cdot}^2 = \frac{4 + 6 + 6}{3} = \frac{16}{3}$$

Example 2 – Forming B and finding its eigenpairs (2/5)

Computing each entry $b_{ij} = -\frac{1}{2}(d_{ij}^2 - \bar{d}_{i\cdot}^2 - \bar{d}_{\cdot j}^2 + \frac{16}{3})$ (or multiplying $-\frac{1}{2}C_3D^{(2)}C_3$ directly):

$$B = -\frac{1}{2}C_3D^{(2)}C_3 = \frac{1}{3}\begin{pmatrix} 4 & -2 & -2 \\ -2 & 10 & -8 \\ -2 & -8 & 10 \end{pmatrix} = \begin{pmatrix} 4/3 & -2/3 & -2/3 \\ -2/3 & 10/3 & -8/3 \\ -2/3 & -8/3 & 10/3 \end{pmatrix}$$

Let us find the eigenvalues and orthonormal eigenvectors of B:

- We always know $\boldsymbol{v}_3 = \frac{1}{\sqrt{3}}(1,1,1)^T$ is an eigenvector with eigenvalue $\lambda_3 = 0$
- By symmetry between points 2 and 3, try $\boldsymbol{v}_1 = \frac{1}{\sqrt{2}}(0,1,-1)^T$:

$$B\boldsymbol{v}_1 = \frac{1}{3\sqrt{2}}\begin{pmatrix} 4 & -2 & -2 \\ -2 & 10 & -8 \\ -2 & -8 & 10 \end{pmatrix}\begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} = \frac{1}{3\sqrt{2}}\begin{pmatrix} 0 \\ 18 \\ -18 \end{pmatrix} = 6\boldsymbol{v}_1 \implies \lambda_1 = 6$$

Example 2 – Second eigenpair and 2D coordinates (3/5)

- Since $\text{tr}(B) = \frac{4+10+10}{3} = 8 = \lambda_1 + \lambda_2 + \lambda_3$ and $\lambda_1 = 6, \lambda_3 = 0$, the remaining eigenvalue is
$$\lambda_2 = 8 - 6 - 0 = 2$$
- The unit eigenvector $\boldsymbol{v}_2$ must be orthogonal to both $\boldsymbol{v}_3 = \frac{1}{\sqrt{3}}(1,1,1)^T$ and $\boldsymbol{v}_1 = \frac{1}{\sqrt{2}}(0,1,-1)^T$, which gives
$$\boldsymbol{v}_2 = \frac{1}{\sqrt{6}} \begin{pmatrix} 2 \\ -1 \\ -1 \end{pmatrix} \qquad \left(\text{check: } B\boldsymbol{v}_2 = \frac{1}{3\sqrt{6}} \begin{pmatrix} 12 \\ -6 \\ -6 \end{pmatrix} = 2\boldsymbol{v}_2 \right)$$

Since $\lambda_1 = 6 > 0$, $\lambda_2 = 2 > 0$ and $\lambda_3 = 0$, $\text{rank}(B) = 2$, so the 3 points embed isometrically in $\mathbb{R}^2$!

Example 2 – Constructing X_2 (4/5)

Using $X_2 = V_2 \Lambda_2^{1/2} = \begin{pmatrix} \sqrt{\lambda_1}\,\boldsymbol{v}_1 & \sqrt{\lambda_2}\,\boldsymbol{v}_2 \end{pmatrix}$ with $\sqrt{\lambda_1} = \sqrt{6}$ and $\sqrt{\lambda_2} = \sqrt{2}$:

$$X_2 = \begin{pmatrix} \sqrt{6} \cdot 0 & \sqrt{2} \cdot \frac{2}{\sqrt{6}} \\ \sqrt{6} \cdot \frac{1}{\sqrt{2}} & \sqrt{2} \cdot \frac{-1}{\sqrt{6}} \\ \sqrt{6} \cdot \frac{-1}{\sqrt{2}} & \sqrt{2} \cdot \frac{-1}{\sqrt{6}} \end{pmatrix} = \begin{pmatrix} 0 & \frac{2}{\sqrt{3}} \\ \sqrt{3} & -\frac{1}{\sqrt{3}} \\ -\sqrt{3} & -\frac{1}{\sqrt{3}} \end{pmatrix} \in \mathcal{M}_{32}(\mathbb{R})$$

- The rows $\boldsymbol{x}_1^T, \boldsymbol{x}_2^T, \boldsymbol{x}_3^T$ are the reconstructed 2D coordinates:

$$\boldsymbol{x}_1 = \begin{pmatrix} 0 \\ 2/\sqrt{3} \end{pmatrix}, \qquad \boldsymbol{x}_2 = \begin{pmatrix} \sqrt{3} \\ -1/\sqrt{3} \end{pmatrix}, \qquad \boldsymbol{x}_3 = \begin{pmatrix} -\sqrt{3} \\ -1/\sqrt{3} \end{pmatrix}$$

- Centred at the origin (column sums are zero):

$$0 + \sqrt{3} - \sqrt{3} = 0, \qquad \frac{2}{\sqrt{3}} - \frac{1}{\sqrt{3}} - \frac{1}{\sqrt{3}} = 0$$

Example 2 – Verifying reconstructed distances (5/5)

Pairwise squared Euclidean distances between the rows $\boldsymbol{x}_1^T, \boldsymbol{x}_2^T, \boldsymbol{x}_3^T$ of X_2:

$$\|\boldsymbol{x}_1 - \boldsymbol{x}_2\|^2 = (0 - \sqrt{3})^2 + \left(\frac{2}{\sqrt{3}} + \frac{1}{\sqrt{3}}\right)^2 = 3 + 3 = 6 = d_{12}^2$$

$$\|\boldsymbol{x}_1 - \boldsymbol{x}_3\|^2 = (0 + \sqrt{3})^2 + \left(\frac{2}{\sqrt{3}} + \frac{1}{\sqrt{3}}\right)^2 = 3 + 3 = 6 = d_{13}^2$$

$$\|\boldsymbol{x}_2 - \boldsymbol{x}_3\|^2 = (\sqrt{3} + \sqrt{3})^2 + 0^2 = (2\sqrt{3})^2 = 12 = d_{23}^2$$

- All pairwise squared distances match $D^{(2)}$ exactly
- The 3 points form an isosceles right triangle centred at the origin in $\mathbb{R}^2$!

Example 3 – 4 points in $\mathbb{R}^3$ forming a pyramid (1/6)

Consider $n = 4$ points in $\mathbb{R}^3$ forming a triangular pyramid (tetrahedron):

- **Base triangle in the xy-plane ($z = 0$)**:

$$\boldsymbol{x}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \qquad \boldsymbol{x}_2 = \begin{pmatrix} -1 \\ 0 \\ 0 \end{pmatrix}, \qquad \boldsymbol{x}_3 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

- **Apex elevated along the z-axis at height $h = 1$**:

$$\boldsymbol{x}_4 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

Pairwise squared Euclidean distances:

- Base edges: $d_{12}^2 = (1 - (-1))^2 = 4, \quad d_{13}^2 = 1^2 + (-1)^2 = 2, \quad d_{23}^2 = 2$
- Lateral edges (apex to base): $d_{14}^2 = 1^2 + 1^2 = 2, \quad d_{24}^2 = 2, \quad d_{34}^2 = 2$

Example 3 – Double-centring $D^{(2)}$ (2/6)

The squared Euclidean distance matrix $D^{(2)} \in \mathcal{M}_4(\mathbb{R})$ is:

$$D^{(2)} = \begin{pmatrix} 0 & 4 & 2 & 2 \\ 4 & 0 & 2 & 2 \\ 2 & 2 & 0 & 2 \\ 2 & 2 & 2 & 0 \end{pmatrix}$$

To compute $B = -\frac{1}{2} C_4 D^{(2)} C_4$ via $b_{ij} = -\frac{1}{2}(d_{ij}^2 - \bar{d}_{i\cdot}^2 - \bar{d}_{\cdot j}^2 + \bar{d}_{\cdot\cdot}^2)$:

- **Row (and column) means**:

$$\bar{d}_{1\cdot}^2 = \frac{0+4+2+2}{4} = 2, \qquad \bar{d}_{2\cdot}^2 = \frac{4+0+2+2}{4} = 2$$

$$\bar{d}_{3\cdot}^2 = \frac{2+2+0+2}{4} = \frac{3}{2}, \qquad \bar{d}_{4\cdot}^2 = \frac{2+2+2+0}{4} = \frac{3}{2}$$

- **Grand mean**:

$$\bar{d}_{\cdot\cdot}^2 = \frac{1}{4}\left(2+2+\frac{3}{2}+\frac{3}{2}\right) = \frac{7}{4}$$

Example 3 – The double-centred Gram matrix B (3/6)

Substituting the means into $b_{ij} = -\frac{1}{2}\left(d_{ij}^2 - \bar{d}_{i\cdot}^2 - \bar{d}_{\cdot j}^2 + \frac{7}{4}\right)$:

$$B = -\frac{1}{2} C_4 D^{(2)} C_4 = \frac{1}{8} \begin{pmatrix} 9 & -7 & -1 & -1 \\ -7 & 9 & -1 & -1 \\ -1 & -1 & 5 & -3 \\ -1 & -1 & -3 & 5 \end{pmatrix}$$

Properties of B:

- **Centring**: each row and column sum is zero ($B\mathbb{1} = \mathbf{0}$):

$$\text{Row 1: } 9 - 7 - 1 - 1 = 0, \qquad \text{Row 3: } -1 - 1 + 5 - 3 = 0$$

- **Trace**:

$$\operatorname{tr}(B) = \frac{9 + 9 + 5 + 5}{8} = \frac{28}{8} = \frac{7}{2} = 3.5$$

Example 3 – Spectral decomposition and dimension (4/6)

Finding the eigenpairs of $B = \frac{1}{8}\begin{pmatrix} 9 & -7 & -1 & -1 \\ -7 & 9 & -1 & -1 \\ -1 & -1 & 5 & -3 \\ -1 & -1 & -3 & 5 \end{pmatrix}$:

- $\boldsymbol{v}_1 = \frac{1}{\sqrt{2}}(1,-1,0,0)^T \implies B\boldsymbol{v}_1 = 2\boldsymbol{v}_1 \implies \lambda_1 = 2$
- $\boldsymbol{v}_2 = \frac{1}{\sqrt{2}}(0,0,-1,1)^T \implies B\boldsymbol{v}_2 = 1\boldsymbol{v}_2 \implies \lambda_2 = 1$
- $\boldsymbol{v}_3 = \frac{1}{2}(1,1,-1,-1)^T \implies B\boldsymbol{v}_3 = \frac{1}{2}\boldsymbol{v}_3 \implies \lambda_3 = \frac{1}{2}$
- $\boldsymbol{v}_4 = \frac{1}{2}(1,1,1,1)^T \implies B\boldsymbol{v}_4 = \boldsymbol{0} \implies \lambda_4 = 0$

Minimum embedding dimension:

- Notice that $\lambda_1 + \lambda_2 + \lambda_3 = 2 + 1 + 0.5 = 3.5 = \text{tr}(B)$
- Exactly 3 eigenvalues are strictly positive: $\lambda_1, \lambda_2, \lambda_3 > 0$
- Hence $\text{rank}(B) = 3 \implies$ the points require a **minimum of 3 dimensions** to embed isometrically!

Example 3 – Recovering 3D coordinates (5/6)

With $r = 3$ and $\Lambda_3 = \mathrm{diag}(2, 1, 1/2)$, the coordinate matrix is:

$$X_3 = V_3 \Lambda_3^{1/2} = \begin{pmatrix} \sqrt{2}\,\boldsymbol{v}_1 & 1\,\boldsymbol{v}_2 & \frac{1}{\sqrt{2}}\,\boldsymbol{v}_3 \end{pmatrix} = \begin{pmatrix} 1 & 0 & \frac{\sqrt{2}}{4} \\ -1 & 0 & \frac{\sqrt{2}}{4} \\ 0 & -\frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{4} \\ 0 & \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{4} \end{pmatrix} \in \mathcal{M}_{43}(\mathbb{R})$$

Geometric verification:

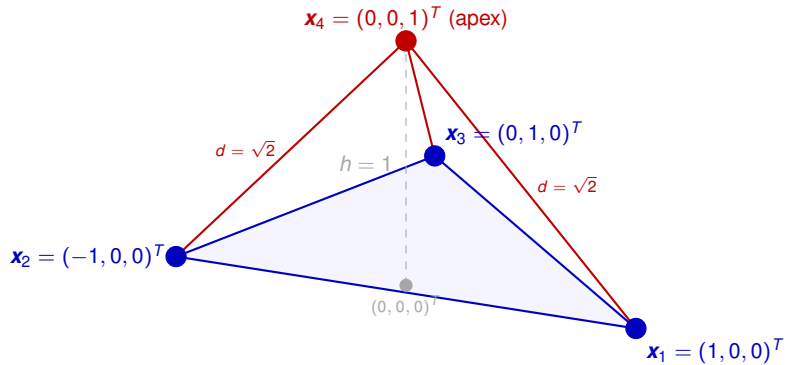
- Column sums are zero: centroid $\bar{\boldsymbol{x}} = \boldsymbol{0}$
- Pairwise squared distances between rows:

$$\|\hat{\boldsymbol{x}}_1 - \hat{\boldsymbol{x}}_2\|^2 = 2^2 + 0^2 + 0^2 = 4 = d_{12}^2$$

$$\|\hat{\boldsymbol{x}}_1 - \hat{\boldsymbol{x}}_3\|^2 = 1^2 + \left(\frac{\sqrt{2}}{2}\right)^2 + \left(\frac{\sqrt{2}}{2}\right)^2 = 1 + \frac{1}{2} + \frac{1}{2} = 2 = d_{13}^2$$

Exact isometry in $\mathbb{R}^3$!

Example 3 – Geometric visualization of the 3D pyramid (6/6)



3D Pyramid: $r = \text{rank}(B) = 3$ positive eigenvalues ($\lambda_1 = 2, \lambda_2 = 1, \lambda_3 = 1/2$)
The positive eigenvalue $\lambda_3 = 1/2$ captures the elevation $h = 1$ of the apex

Example 3b – Flattening the pyramid into the base plane (1/6)

Now suppose the apex $\boldsymbol{x}_4$ is flattened down into the base plane ($z = 0$):

- The apex descends from $(0, 0, 1)^T$ to the origin: $\boldsymbol{x}_4 = (0, 0, 0)^T$
- The 4 points now lie in the same 2D plane:

$$\boldsymbol{x}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad \boldsymbol{x}_2 = \begin{pmatrix} -1 \\ 0 \end{pmatrix}, \qquad \boldsymbol{x}_3 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \qquad \boldsymbol{x}_4 = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

New pairwise squared distances:

- Base distances are unchanged: $d_{12}^2 = 4, \quad d_{13}^2 = 2, \quad d_{23}^2 = 2$
- Distances from $\boldsymbol{x}_4 = \mathbf{0}$ become:

$$d_{14}^2 = 1^2 + 0^2 = 1, \qquad d_{24}^2 = (-1)^2 + 0^2 = 1, \qquad d_{34}^2 = 0^2 + 1^2 = 1$$

Example 3b – Double-centring $D^{(2)}_{\text{flat}}$ (2/6)

The squared distance matrix of the flattened configuration is:

$$D^{(2)}_{\text{flat}} = \begin{pmatrix} 0 & 4 & 2 & 1 \\ 4 & 0 & 2 & 1 \\ 2 & 2 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{pmatrix}$$

To compute $B_{\text{flat}} = -\frac{1}{2} C_4 D^{(2)}_{\text{flat}} C_4$ via $b_{ij} = -\frac{1}{2}(d^2_{ij} - \bar{d}^2_{i\cdot} - \bar{d}^2_{\cdot j} + \bar{d}^2_{\cdot\cdot})$:

- **Row (and column) means**:

$$\bar{d}^2_{1\cdot} = \frac{0+4+2+1}{4} = \frac{7}{4}, \qquad \bar{d}^2_{2\cdot} = \frac{4+0+2+1}{4} = \frac{7}{4}$$

$$\bar{d}^2_{3\cdot} = \frac{2+2+0+1}{4} = \frac{5}{4}, \qquad \bar{d}^2_{4\cdot} = \frac{1+1+1+0}{4} = \frac{3}{4}$$

- **Grand mean**:

$$\bar{d}^2_{\cdot\cdot} = \frac{1}{4}\left(\frac{7}{4} + \frac{7}{4} + \frac{5}{4} + \frac{3}{4}\right) = \frac{22}{16} = \frac{11}{8}$$

Example 3b – The double-centred Gram matrix B_{flat} (3/6)

Substituting the means into $b_{ij} = -\frac{1}{2}\left(d_{ij}^2 - \bar{d}_{i\cdot}^2 - \bar{d}_{\cdot j}^2 + \frac{11}{8}\right)$:

$$B_{\text{flat}} = -\frac{1}{2} C_4 D_{\text{flat}}^{(2)} C_4 = \frac{1}{16} \begin{pmatrix} 17 & -15 & -3 & 1 \\ -15 & 17 & -3 & 1 \\ -3 & -3 & 9 & -3 \\ 1 & 1 & -3 & 1 \end{pmatrix}$$

Properties of B_{flat}:

- **Centring**: each row and column sum is zero ($B_{\text{flat}}\mathbb{1} = \mathbf{0}$):

$$\text{Row 1: } 17 - 15 - 3 + 1 = 0, \qquad \text{Row 3: } -3 - 3 + 9 - 3 = 0$$

- **Trace**:

$$\operatorname{tr}(B_{\text{flat}}) = \frac{17 + 17 + 9 + 1}{16} = \frac{44}{16} = \frac{11}{4} = 2.75$$

Example 3b – Dimensional collapse from 3D to 2D (4/6)

Finding the eigenpairs of B_{flat}:

- $\boldsymbol{v}_1 = \frac{1}{\sqrt{2}}(1, -1, 0, 0)^T \implies B_{\text{flat}}\boldsymbol{v}_1 = 2\boldsymbol{v}_1 \implies \lambda_1 = 2$
- $\boldsymbol{v}_2 = \frac{1}{2\sqrt{3}}(1, 1, -3, 1)^T \implies B_{\text{flat}}\boldsymbol{v}_2 = \frac{3}{4}\boldsymbol{v}_2 \implies \lambda_2 = \frac{3}{4} = 0.75$
- Trace check: $\lambda_1 + \lambda_2 = 2 + 0.75 = 2.75 = \text{tr}(B_{\text{flat}})$
- Therefore, the remaining eigenvalues are:

$$\lambda_3 = 0, \qquad \lambda_4 = 0$$

Comparison of eigenvalues and dimension:

Configuration	λ_1	λ_2	λ_3	rank(B)
3D Pyramid ($h = 1$)	2	1	**1/2**	**3**
Flattened ($h = 0$)	2	3/4	**0**	**2**

The out-of-plane eigenvalue has collapsed: $\lambda_3 = 1/2 \longrightarrow 0$!

Example 3b – Recovering 2D coordinates (5/6)

With $r = 2$ and $\Lambda_2 = \mathrm{diag}(2, 3/4)$, the coordinate matrix is:

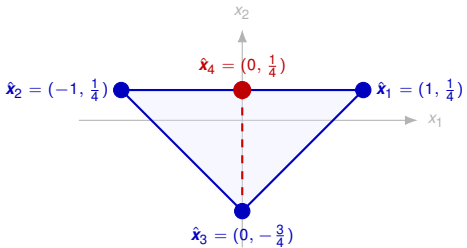
$$X_2 = V_2 \Lambda_2^{1/2} = \begin{pmatrix} \sqrt{2}\,\boldsymbol{v}_1 & \frac{\sqrt{3}}{2}\,\boldsymbol{v}_2 \end{pmatrix} = \begin{pmatrix} 1 & \frac{1}{4} \\ -1 & \frac{1}{4} \\ 0 & -\frac{3}{4} \\ 0 & \frac{1}{4} \end{pmatrix} \in \mathcal{M}_{42}(\mathbb{R})$$

Verification of pairwise distances:

- $\|\hat{\boldsymbol{x}}_1 - \hat{\boldsymbol{x}}_2\|^2 = 2^2 + 0^2 = 4 = d_{12}^2$
- $\|\hat{\boldsymbol{x}}_1 - \hat{\boldsymbol{x}}_3\|^2 = 1^2 + (1/4 - (-3/4))^2 = 1 + 1 = 2 = d_{13}^2$
- $\|\hat{\boldsymbol{x}}_1 - \hat{\boldsymbol{x}}_4\|^2 = 1^2 + (1/4 - 1/4)^2 = 1 = d_{14}^2$
- $\|\hat{\boldsymbol{x}}_3 - \hat{\boldsymbol{x}}_4\|^2 = 0^2 + (-3/4 - 1/4)^2 = 1 = d_{34}^2$

All pairwise distances are preserved perfectly in $\mathbb{R}^2$!

Example 3b – Comparative geometric visualization (6/6)



Flattened configuration in $\mathbb{R}^2$

Core takeaways:

- **Eigenvalues diagnose intrinsic dimension**: The number of strictly positive eigenvalues of B gives the minimal Euclidean dimension $r = \text{rank}(B)$
- **Continuous collapse**: As the height of the pyramid decreases $h \to 0$, the third eigenvalue continuously shrinks $\lambda_3(h) \to 0$
- When points become coplanar, $\lambda_3 = 0$ exactly, and the embedding dimension drops from 3 to 2

Motivation: from distances back to coordinates

Gram matrices and Euclidean distance matrices

The centring matrix as an orthogonal projector

Classical multidimensional scaling (MDS)

Algebraic duality between PCA and classical MDS

Algebraic duality between PCA and classical MDS

Equivalence via the singular value decomposition

Application in R: reconstructing maps from distances

Two $n \times p$ products: $\widetilde{X}^T\widetilde{X}$ versus $\widetilde{X}\widetilde{X}^T$

Suppose we have a data matrix $X \in \mathcal{M}_{np}(\mathbb{R})$ and its column-centred data matrix $\widetilde{X} = C_n X \in \mathcal{M}_{np}(\mathbb{R})$. From $\widetilde{X}$, we can form two symmetric positive semi-definite matrices:

Feature scatter / covariance (PCA, Unit U05)

$$\widetilde{X}^T\widetilde{X} \in \mathcal{M}_p(\mathbb{R}) \quad \left(\text{or } S_X = \frac{1}{n-1}\widetilde{X}^T\widetilde{X}\right)$$

- Size: $p \times p$ (features $\times$ features)
- (j, ℓ) entry: inner product between centred column j and centred column ℓ
- Diagonalising $\widetilde{X}^T\widetilde{X}$ gives the principal directions (loadings) in $\mathbb{R}^p$

Centred Gram matrix (Classical MDS, Unit U07)

$$B = \widetilde{X}\widetilde{X}^T = -\frac{1}{2}C_n D^{(2)} C_n \in \mathcal{M}_n(\mathbb{R})$$

- Size: $n \times n$ (observations $\times$ observations)
- (i, j) entry: inner product $\tilde{\boldsymbol{x}}_i^T \tilde{\boldsymbol{x}}_j$ between centred observation i and centred observation j
- Diagonalising $\widetilde{X}\widetilde{X}^T$ gives the principal coordinates in $\mathbb{R}^n$

Connecting PCA and classical MDS via the SVD

Let $\widetilde{X} = C_n X \in \mathcal{M}_{np}(\mathbb{R})$ be the column-centred data matrix of rank $r \leq \min(n-1, p) < n$, and write its singular value decomposition (Unit U04):

$$\widetilde{X} = U_r \Sigma_r V_r^T$$

where:

- $U_r = [\boldsymbol{u}_1 \ \cdots \ \boldsymbol{u}_r] \in \mathcal{M}_{nr}(\mathbb{R})$ has orthonormal columns ($U_r^T U_r = \mathbb{I}_r$)
- $\Sigma_r = \text{diag}(\sigma_1, \ldots, \sigma_r) \in \mathcal{M}_r(\mathbb{R})$ contains the positive singular values $\sigma_1 \geq \cdots \geq \sigma_r > 0$
- $V_r = [\boldsymbol{v}_1 \ \cdots \ \boldsymbol{v}_r] \in \mathcal{M}_{pr}(\mathbb{R})$ has orthonormal columns ($V_r^T V_r = \mathbb{I}_r$)

Now express both $\widetilde{X}^T \widetilde{X}$ and $B = \widetilde{X}\widetilde{X}^T$ in terms of the SVD of $\widetilde{X}$:

$$\widetilde{X}^T \widetilde{X} = V_r \Sigma_r^2 V_r^T \in \mathcal{M}_p(\mathbb{R}) \qquad \text{and} \qquad B = \widetilde{X}\widetilde{X}^T = U_r \Sigma_r^2 U_r^T \in \mathcal{M}_n(\mathbb{R})$$

Duality theorem: PCA scores equal classical MDS coordinates

Theorem 106 (Equivalence of PCA and Euclidean classical MDS)

Let $X \in \mathcal{M}_{np}(\mathbb{R})$ be a data matrix with column-centred data matrix $\widetilde{X} = C_n X$ of rank $r \leq \min(n-1, p) < n$ having singular value decomposition $\widetilde{X} = U_r \Sigma_r V_r^T$, and let $D = [\|\boldsymbol{x}_i - \boldsymbol{x}_j\|] \in \mathcal{M}_n(\mathbb{R})$ be the Euclidean distance matrix between the rows of X. Then for any $k \leq r$:

1. *The non-zero eigenvalues of $\widetilde{X}^T \widetilde{X} \in \mathcal{M}_p(\mathbb{R})$ and $B = -\frac{1}{2} C_n D^{(2)} C_n = \widetilde{X} \widetilde{X}^T \in \mathcal{M}_n(\mathbb{R})$ are identical:*

$$\lambda_i = \sigma_i^2 > 0, \qquad i = 1, \ldots, r$$

2. *The k-dimensional principal component score matrix $Y_k = \widetilde{X} V_k \in \mathcal{M}_{nk}(\mathbb{R})$ from PCA and the k-dimensional coordinate matrix $X_k = U_k \Lambda_k^{1/2} \in \mathcal{M}_{nk}(\mathbb{R})$ from classical MDS on D are **identical** (up to sign choices of the eigenvectors):*

$$Y_k = \widetilde{X} V_k = U_k \Sigma_k = U_k \Lambda_k^{1/2} = X_k$$

Verifying PCA and classical MDS equivalence in R

Let us first verify Theorem 106 numerically in R on a synthetic 5×3 centred data matrix $\widetilde{X}$:

```
set.seed(2740)
X_tilde = scale(matrix(rnorm(15), nrow = 5, ncol = 3), center = TRUE, scale
pca_scores = prcomp(X_tilde, center = FALSE)$x[, 1:2]
mds_coords = cmdscale(dist(X_tilde), k = 2)
# Compare absolute values (eigenvectors are defined up to a sign +/- 1)
max(abs(abs(pca_scores) - abs(mds_coords)))

## [1] 2.442491e-15
```

Up to machine precision ($\approx 10^{-16}$) and column signs, the PCA scores `prcomp(X_tilde)$x[,1:2]` and the classical MDS coordinates `cmdscale(dist(X_tilde), k=2)` are identical!

Computational and practical payoff of PCA–MDS duality

Why is Theorem 106 so useful in practice?

1. **Tall-and-skinny data ($n \gg p$, many observations, few features)**:
   - $\widetilde{X}^T\widetilde{X}$ is a small $p \times p$ matrix, whereas $\widetilde{X}\widetilde{X}^T$ is a huge $n \times n$ matrix
   - Computing PCA via $\widetilde{X}^T\widetilde{X}$ costs $O(np^2 + p^3)$, which is much faster than $O(n^3)$ for $\widetilde{X}\widetilde{X}^T$
2. **Short-and-wide data ($p \gg n$, few observations, huge number of features)**:
   - E.g. gene expression ($n = 50$ patients, $p = 20{,}000$ genes) or high-resolution images ($n = 100$ faces, $p = 10^6$ pixels)
   - $\widetilde{X}^T\widetilde{X}$ is $20{,}000 \times 20{,}000$, whereas the Gram matrix $B = \widetilde{X}\widetilde{X}^T$ is only 50×50!
   - Diagonalising the $n \times n$ Gram matrix $B = \widetilde{X}\widetilde{X}^T$ (classical MDS) gives the exact PCA scores in $O(n^2p + n^3)$ time!
3. **Coordinate-free and kernel extensions**:
   - Classical MDS works when X is unknown and only distances D are given
   - Replacing $\boldsymbol{x}_i^T\boldsymbol{x}_j$ by a general similarity kernel $k(\boldsymbol{x}_i, \boldsymbol{x}_j)$ leads directly to **kernel PCA**!

Algebraic duality between PCA and classical MDS

Equivalence via the singular value decomposition

Application in R: reconstructing maps from distances

Geographical example: Canadian cities distance matrix

Now suppose we are only given approximate great-circle distances (in kilometres) between $n = 8$ Canadian cities, with **no latitude or longitude coordinates**:

```
cities = c("Victoria", "Vancouver", "Calgary", "Regina",
           "Winnipeg", "Toronto", "Montreal", "Halifax")
D_can = matrix(c(
     0,   90,  725, 1215, 1690, 3390, 3760, 4490,
    90,    0,  675, 1165, 1640, 3355, 3685, 4430,
   725,  675,    0,  505,  990, 2705, 3020, 3770,
  1215, 1165,  505,    0,  500, 2030, 2360, 3125,
  1690, 1640,  990,  500,    0, 1515, 1825, 2590,
  3390, 3355, 2705, 2030, 1515,    0,  505, 1265,
  3760, 3685, 3020, 2360, 1825,  505,    0,  790,
  4490, 4430, 3770, 3125, 2590, 1265,  790,    0
), nrow = 8, byrow = TRUE, dimnames = list(cities, cities))
```

Step-by-step manual classical MDS vs `cmdscale`

Let us implement the classical MDS algorithm directly from $B = -\frac{1}{2} C_n D^{(2)} C_n$ and compare with R's built-in `cmdscale` function:

```
n = nrow(D_can)
C_n = diag(n) - matrix(1, n, n) / n
B = -0.5 * C_n %*% (D_can^2) %*% C_n
ev = eigen(B, symmetric = TRUE)
round(ev$values, 1)
```

```
## [1] 20985217.2   224941.6    67722.5    17544.1     3459.2        0.0    -
## [8]  -482729.4
```

```
X_manual = ev$vectors[, 1:2] %*% diag(sqrt(ev$values[1:2]))
rownames(X_manual) = cities
mds_builtin = cmdscale(as.dist(D_can), k = 2, eig = TRUE)
max(abs(abs(X_manual) - abs(mds_builtin$points)))
```

```
## [1] 0
```

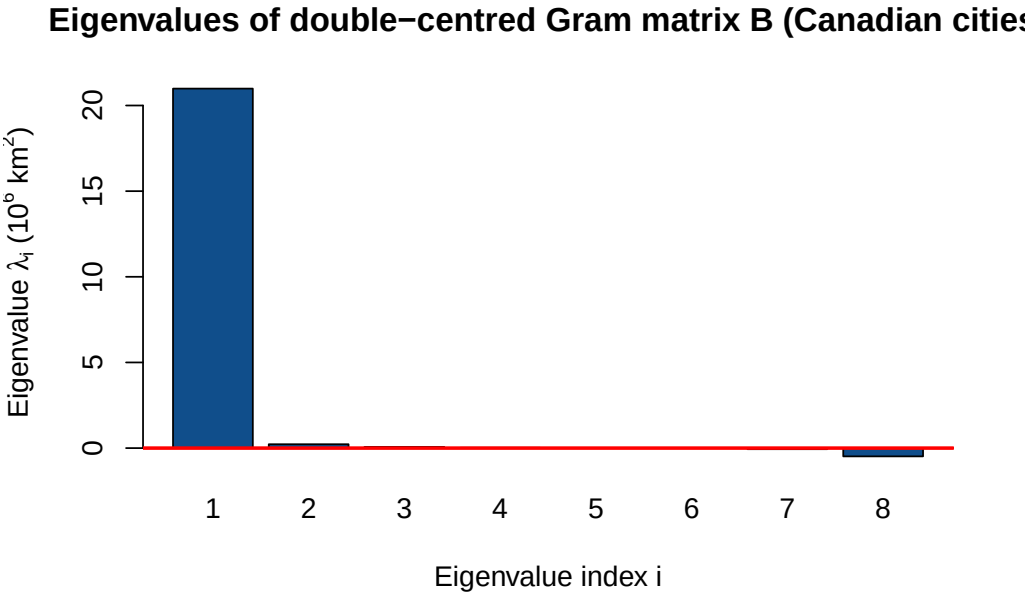
Eigenvalue spectrum and goodness of fit

Notice how the first two eigenvalues dominate the spectrum of B:

```
gof_top2 = sum(ev$values[1:2]) / sum(ev$values[ev$values > 0])
gof_1d   = ev$values[1] / sum(ev$values[ev$values > 0])
c(GOF_1D = round(gof_1d, 4), GOF_2D = round(gof_top2, 4))

## GOF_1D GOF_2D
## 0.9853 0.9958
```

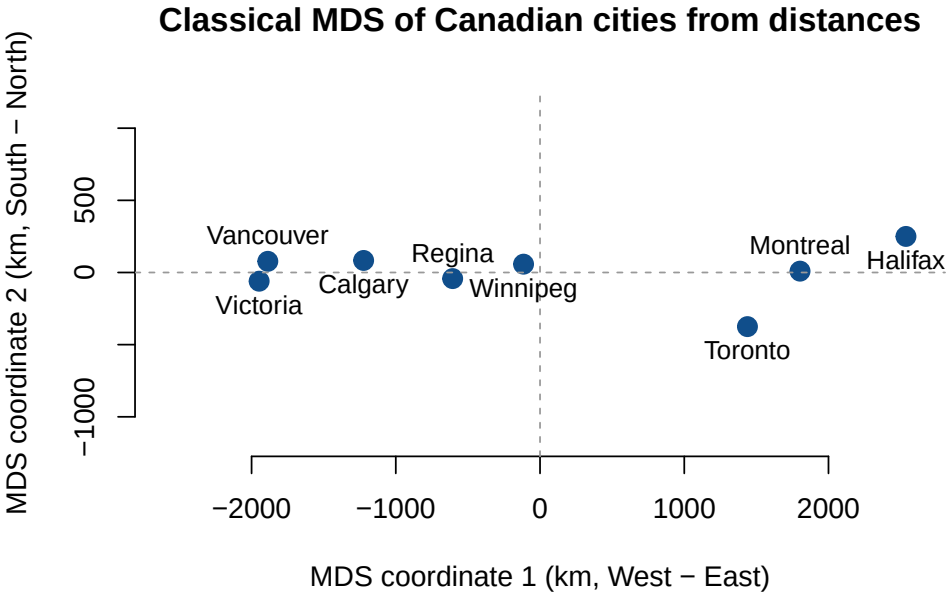
The first dimension alone (East–West axis across Canada) explains 98.5% of the positive trace, and 2 dimensions explain 99.58%!


Eigenvalues of double−centred Gram matrix B (Canadian cities
Eigenvalue λ_i (10^6 km^2)
0
5
10
15
20
1
2
3
4
5
6
7
8
Eigenvalue index i

Plotting the reconstructed 2D map of Canadian cities

Recall that classical MDS only determines coordinates up to orthogonal transformations (rotations and reflections). We plot the raw MDS coordinates and then flip signs if needed so West is left and North is up:

```
coords = X_manual
# Ensure West < 0 and Winnipeg is North of Toronto
if (coords["Vancouver", 1] > 0) coords[, 1] = -coords[, 1]
if (coords["Winnipeg", 2] < coords["Toronto", 2]) coords[, 2] = -coords[, 2]
plot(coords[, 1], coords[, 2],
     pch = 19, cex = 1.5, col = "dodgerblue4",
     asp = 1, bty = "n", xlim = c(-2600, 2600), ylim = c(-900, 900),
     xlab = "MDS coordinate 1 (km, West - East)",
     ylab = "MDS coordinate 2 (km, South - North)",
     main = "Classical MDS of Canadian cities from distances")
abline(h = 0, v = 0, lty = 2, col = "grey60")
```


Classical MDS of Canadian cities from distances
MDS coordinate 2 (km, South − North)
500
0
−1000
Vancouver
Victoria
Calgary
Regina
Winnipeg
Montreal
Halifax
Toronto
−2000
−1000
0
1000
2000
MDS coordinate 1 (km, West − East)

What is wrong with this map?

Look closely at the reconstructed map on the previous slide

- **Halifax** has $y \approx +248$ km, while **Victoria** has $y \approx -68$ km
- Halifax appears to be far north of Victoria
- In reality:
  - Victoria is at latitude 48.4° N
  - Halifax is at latitude 44.6° N

  Halifax is actually **south** of Victoria
- **What happened?**
  - Classical MDS recovers coordinates only up to an orthogonal transformation
  - Flipping signs only resolved axis reflections (± 1 along the axes)
  - The first MDS axis aligns with the direction of maximal variance across Canada, which is tilted relative to lines of constant latitude
  - In short, we did not rotate the configuration as we should have, so the orientation is a little off

Deducing the rotation from Victoria and Halifax

Coordinates of the westernmost and easternmost cities:

- **Victoria**: $\boldsymbol{x}_{\text{Vic}} \approx \begin{pmatrix} -1945.8 \\ -68.5 \end{pmatrix}$ km, **Halifax**: $\boldsymbol{x}_{\text{Hal}} \approx \begin{pmatrix} 2538.5 \\ 247.5 \end{pmatrix}$ km

The displacement vector from Victoria to Halifax is:

$$\Delta \boldsymbol{x} = \boldsymbol{x}_{\text{Hal}} - \boldsymbol{x}_{\text{Vic}} = \begin{pmatrix} 2538.5 - (-1945.8) \\ 247.5 - (-68.5) \end{pmatrix} = \begin{pmatrix} 4484.3 \\ 316.0 \end{pmatrix} \text{ km}$$

which makes an angle with the horizontal axis of:

$$\theta = \arctan\left(\frac{316.0}{4484.3}\right) \approx 0.0704 \text{ rad} \approx 4.03^\circ$$

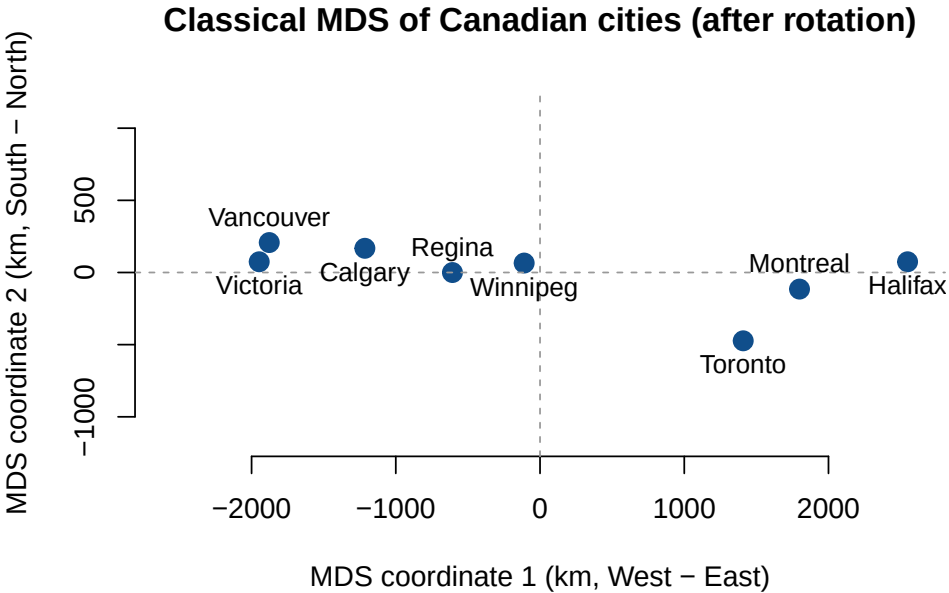
To align this West–East baseline horizontally, we rotate clockwise by $\theta \approx 4.03^\circ$ using the rotation matrix:

$$R_{-\theta} = \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix}, \qquad X_{\text{rot}} = X R_{-\theta}^T$$

Plotting the Canadian cities map after rotation

We apply the rotation matrix $R_{-\theta}$ to `coords` and plot the rotated coordinates:

```
theta = atan2(coords["Halifax", 2] - coords["Victoria", 2],
              coords["Halifax", 1] - coords["Victoria", 1])
R_rot = matrix(c(cos(-theta), -sin(-theta),
                 sin(-theta),  cos(-theta)),
               nrow = 2, byrow = TRUE)
coords_rot = coords %*% t(R_rot)
plot(coords_rot[, 1], coords_rot[, 2],
     pch = 19, cex = 1.5, col = "dodgerblue4",
     asp = 1, bty = "n", xlim = c(-2600, 2600), ylim = c(-900, 900),
     xlab = "MDS coordinate 1 (km, West - East)",
     ylab = "MDS coordinate 2 (km, South - North)",
     main = "Classical MDS of Canadian cities (after rotation)")
abline(h = 0, v = 0, lty = 2, col = "grey60")
```


Classical MDS of Canadian cities (after rotation)
MDS coordinate 2 (km, South − North)
500
0
−1000
Vancouver
Victoria
Calgary
Regina
Winnipeg
Montreal
Halifax
Toronto
−2000
−1000
0
1000
2000
MDS coordinate 1 (km, West − East)

One remaining underlying issue: Earth's curvature

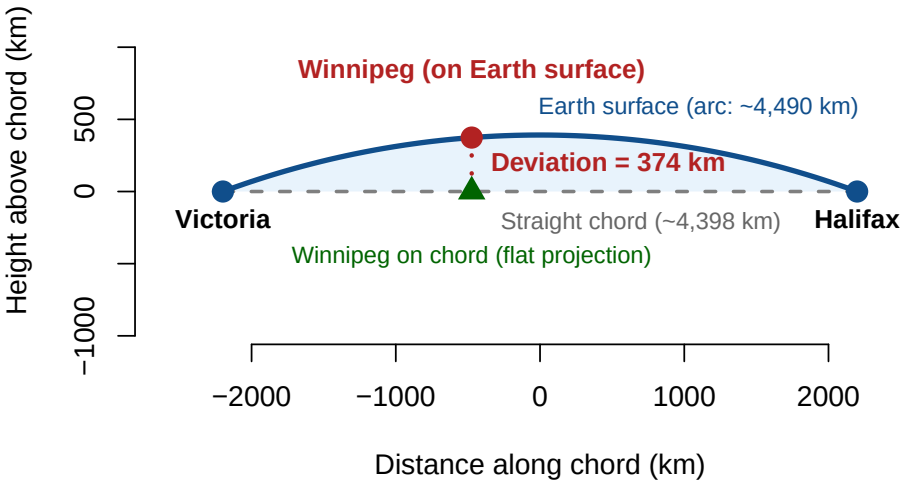
Canada is too large and the Earth is too round..

- Distances D are **great-circle distances** measured along the spherical surface of the Earth ($R \approx 6,371$ km), not straight Euclidean distances in a flat 2D plane
- Great-circle arc between Victoria and Halifac spans $s \approx 4,490$ km, subtending a central angle:

$$\alpha = \frac{s}{R} = \frac{4,490}{6,371} \approx 0.7048 \text{ rad} \approx 40.4^\circ$$

- Straight chord through the interior of the Earth is $2R\sin(\alpha/2) \approx 4,398$ km (92 km shorter than the surface distance)
- Spherical surface bulges up to $R(1 - \cos(\alpha/2)) \approx 391$ km above this chord
- In a flat 2D projection, cities are compressed toward the chord: **Winnipeg** on the Earth's surface deviates by ≈ 374 km from the straight segment connecting Victoria and Halifax

Earth curvature: Victoria to Halifax (true 1:1 scale)



Summary of Unit U07

- **Gram matrix**: for $X \in \mathcal{M}_{np}(\mathbb{R})$, $G = XX^T \in \mathcal{M}_n(\mathbb{R})$ has entries $g_{ij} = \boldsymbol{x}_i^T \boldsymbol{x}_j$, is symmetric positive semi-definite and satisfies $\operatorname{rank}(G) = \operatorname{rank}(X)$
- **From Gram to squared distances**: with $\boldsymbol{g} = \operatorname{diag}(G)$,

$$D^{(2)} = \boldsymbol{g}\mathbb{1}^T - 2G + \mathbb{1}\boldsymbol{g}^T$$

- **Centring matrix**: $C_n = \mathbb{I}_n - \frac{1}{n}\mathbb{1}\mathbb{1}^T$ is the symmetric, idempotent orthogonal projection matrix onto $\operatorname{span}(\mathbb{1})^\perp$ with $\operatorname{rank}(C_n) = \operatorname{tr}(C_n) = n - 1$ and $C_n\mathbb{1} = \mathbf{0}$
- **Double centring and classical MDS**: $B = -\frac{1}{2} C_n D^{(2)} C_n = \widetilde{X}\widetilde{X}^T$ recovers the centred Gram matrix from distances alone, and $X_k = V_k \Lambda_k^{1/2}$ recovers k-dimensional coordinates
- **PCA–MDS duality**: via the SVD $\widetilde{X} = U\Sigma V^T$, classical MDS on Euclidean distances of X produces the exact same low-dimensional coordinates $U_k\Sigma_k$ as PCA on X